

A Review on Integrating Autonomy Into System of Systems: Challenges and Research Directions

MOHAMMADREZA TORKJAZI  (Member, IEEE) AND ALI K. RAZ  (Senior Member, IEEE)

Department of Systems Engineering and Operations Research, George Mason University, Fairfax, VA 22030 USA

CORRESPONDING AUTHOR: MOHAMMADREZA TORKJAZI (e-mail: mtorkjaz@gmu.edu).

ABSTRACT Artificial intelligence and machine learning (AI/ML) technologies convert conventional engineered systems into autonomous systems that are capable of performing tasks in their operational environment with limited to no human involvement. These technologies can reduce demands for human workload while enabling a suite of new capabilities such as autonomous vehicles and smart cities. However, a major challenge is the integration of these autonomous systems into a system of systems (SoSs), essentially resulting in a system of autonomous systems (SoASs). SoASs are fraught with new challenges that compound issues from the founding domains of SoS, AI/ML, and autonomous systems. To understand the new set of challenges for SoAS, this article conducts an extensive review of both theoretical and application-based literature published in the founding domains. The goal is to examine how individual challenges in each domain intersect and exacerbate when multiple independent systems with AI/ML are integrated into an SoAS. A particular emphasis is placed on highlighting how interactions across these domains manifest at the SoAS level. As a result, four overarching challenges for SoASs are identified that must be addressed by systems engineers to ensure a successful realization of the SoAS in the future: 1) SoAS foundation; 2) emergence, safety, and performance; 3) architecture and integration; and 4) test and evaluation. Each challenge is comprehensively examined in the three founding domains by discussing domain-specific state-of-the-art methods and tools that different engineering disciplines have proposed to address. For each challenge, we also investigated how the existing tools and methods apply to addressing the challenge for SoAS and highlighted the remaining gaps that still need to be addressed. Furthermore, this article identifies systems engineering research needs for improving SoAS foundations, analysis of autonomy impacts, and enabling SoAS architecture, integration, and evaluation methods. Conducting research studies in these fields will improve systems engineering practices for a successful and effective realization of SoAS.

INDEX TERMS Autonomy integration, SoS challenges, SoS engineering, system of autonomous systems (SoASs), system of systems (SoSs).

I. INTRODUCTION

Recent advances in artificial intelligence and machine learning (AI/ML) have enabled autonomous systems that intelligently interact with their environment to achieve predefined goals, with varying degrees of human intervention. These systems are finding unique and novel solutions in different areas of society, including but not limited to, transportation, energy production, agriculture, and national security. Although this new class of systems is currently being designed and developed as independent systems, they will inevitably be

integrated into a System of Systems (SoSs) [1], [2] where the result becomes a System of Autonomous Systems (SoASs) [3] that leads to innovations such as smart grid [4], smart waste management system, and smart houses [5].

The SoAS is essentially an extension of the SoS concept where constituent systems are not only managerially and operationally independent, but can also be autonomous. An SoAS can be defined as a purposefully integrated collection of multiple systems that individually employ some form of AI/ML to achieve their independent purposes but come

together to enable capabilities that cannot be provided by each system alone. These capabilities result from the integration and cooperation between systems and go beyond the simple accumulation of each system's capabilities.

Each system may decide and act based on its limited observations and interaction with the environment, along with taking advantage of the collective information and decision-making enabled by the cooperation of other systems [6]. Systems can leverage AI cooperation algorithms to provide the new SoAS capabilities and accomplish missions that are hazardous to human resources and need a high-level computational or cognitive effort. A few examples of these missions are simultaneously performing multidomain military operations in several geographical locations [7], [8], doing persistent coastal surveillance [9], searching for and rescuing survivors of flooding disasters [10], spraying agricultural pesticides [11], [12], and providing more efficient goods distribution and city logistics in crowded metropolitan areas [13], [14].

While the integration of autonomous systems into SoSs is inevitable [5], there are major systems engineering challenges that must be recognized and addressed for the successful realization of SoAS. To understand these challenges, there is a need to investigate the technical foundation of the SoAS. This foundation lies at the intersection of well-established disciplines of SoS, AI/ML, and autonomous systems. These three domains will be referred to as the SoAS founding domains in the rest of this article. Researchers in a variety of engineering disciplines have identified several challenges for these three independent domains which not only compound together for SoAS but also lead to new challenges specific to SoAS. The literature, however, lacks a comprehensive study of how the challenges in founding domains impact SoAS and what new challenges emerge at the SoAS level as a result of integrating independent autonomous systems.

The contributions of this article are twofold. First, we present an interdisciplinary literature review to thoroughly examine how individual systems engineering challenges in each founding domain intersect and amplify when multiple independent systems with AI/ML are integrated into an SoAS. The focus of this review is on identifying challenges that arise with the integration of systems engineered with AI/ML in SoAS as opposed to reviewing the development or use of AI/ML for SoAS engineering. This review was conducted by examining a variety of theoretical studies and real-world application case studies in various engineering disciplines. Upon reviewing 195 published articles, we concluded four major challenges for SoAS. These challenges are concerned with understanding SoAS foundations, the impact of SoAS emergence on safety and performance, crucial considerations in SoAS architecture and integration, and finally, the methods and frameworks to be taken in SoAS test and evaluation. Our second contribution is providing four potential research directions to guide systems engineers in addressing the identified challenges. These research directions include establishing a shared ontology for improving the SoAS foundations, updating safety standards

and employing innovative analysis methods for SoAS, creating an architecting and integration methodology, and using new methods for testing and evaluation of SoAS emergent behaviors.

The rest of this article is organized as follows: Section II provides the research methodology used in conducting the literature review across AI/ML, autonomous systems, and SoS domains from a Systems Engineering standpoint. A brief review of the technical foundations of AI/ML, autonomous systems, and SoS along with their related particular challenges is provided in Section III. Section IV discusses how the challenges in each domain intersect and result in the four major SoAS challenges. Furthermore, four potential research directions are offered in Section V to guide future research studies in further improving systems engineering practices for SoAS. Finally, Section VI summarizes this review study.

II. RESEARCH METHODOLOGY

The findings described in this article are based on reviewing published systems engineering-related literature in AI/ML, autonomous systems, and SoS domains over the past 25 years. Fig. 1(a) lists the publishers, the number of referred studies from each publisher, and the major referred sources in this review. The concept of SoAS in the related literature has been described using a variety of terms depending on different engineering disciplines and various research topics in both military and civilian domains. Therefore, papers in this review were collected by using keywords including “system of systems,” “autonomous systems,” “autonomous system of systems,” “levels of autonomy,” “multi-UAV systems,” “multi-agent systems,” “mission planning,” “SoS challenges,” and “distributed architecture.” These terms were looked up on various databases including Google Scholar, IEEE Xplore, Wiley Online Library, Association for Computing Machinery Digital Library, ScienceDirect, etc. This resulted in the selection of 195 papers that were focused on our study goals.

The identified papers cover various fields of study that are directly tied to AI/ML, autonomous systems, and SoS as shown in Fig. 1(b). The fields also demonstrate that papers were obtained from various engineering disciplines including robotics, electrical engineering, systems engineering, operations research, aerospace, computer science, and information systems. The top two categories of papers are related to the benefits and challenges of autonomy and AI cooperation algorithms.

With respect to the application domains of articles, 96 out of 195 papers were theoretical studies and the rest were focused on applied research. The theoretical papers are mostly concerned with definitions, challenges, characteristics, guidelines, methodologies, and algorithm development, while the applied research papers address missions in both civilian and military application domains with an emphasis on surveillance, aerospace, combat, maritime, navigation, and firefighting. It is evident that research in various domains is

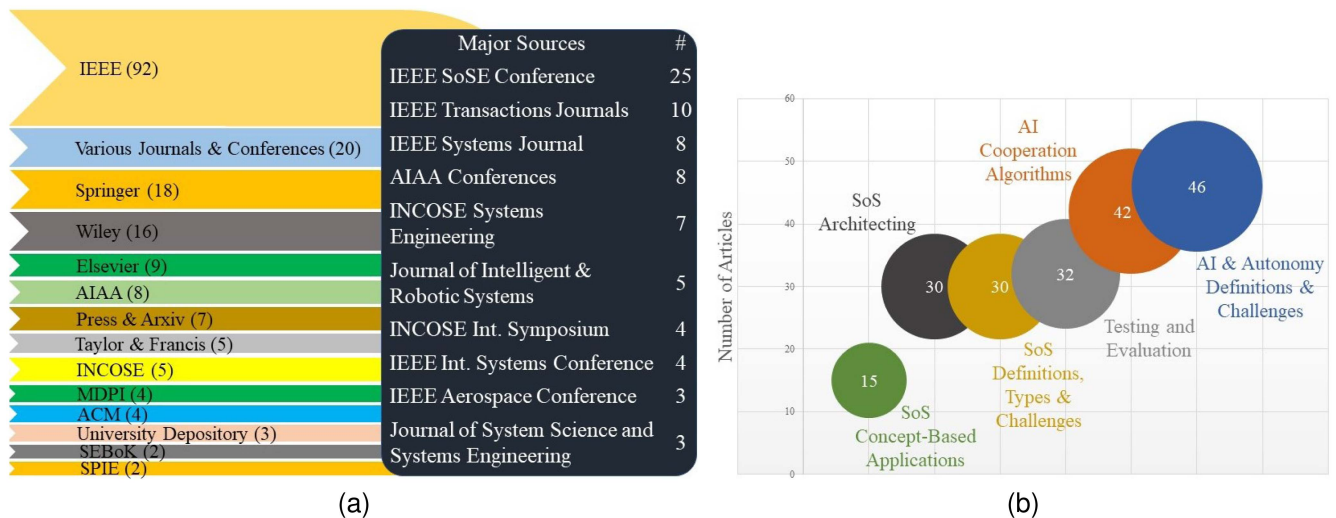


FIGURE 1. (a) Sources of the reviewed literature. (b) Fields of the reviewed literature.

demonstrating a strong interest in finding solutions for integrating autonomous systems into an SoS.

The main objective of this review study is to identify major systems engineering challenges for a successful realization of SoAS. To achieve this objective, we first review the literature to identify particular challenges in each SoAS founding domain. Then, we examine how these challenges can exacerbate and manifest in the SoAS context. Next, we assess the applicability and limitations of the already-developed methods and tools in each founding domain to address the challenges. Subsequently, we propose four potential research directions to further improve the systems engineering practices in addressing SoAS challenges.

III. TECHNICAL FOUNDATIONS

This section introduces the founding domains of SoAS and their particular challenges for systems engineers. The main objective is to get familiar with existing challenges in each domain. This knowledge assists us in understanding the inter-relationships among these challenges and how they manifest at the SoAS level.

A. SYSTEM OF AUTONOMOUS SYSTEMS

According to the INCOSE Systems Engineering Vision 2035 [5], a more interconnected world as a result of technological advancements like AI/ML necessitates new system solutions to handle emerging challenges. Systems equipped with AI/ML technologies can convert an SoS to SoAS. An SoAS can be composed of heterogeneous systems, each having a different level of autonomy (LoA), where LoA refers to the level of autonomous capabilities offered by the system depending on its underlying AI/ML technology [15], [16]. As a result, an SoAS is formed by the three founding domains of AI/ML, autonomous systems, and SoS.

Although SoASs are able to address the needs of the new interconnected world by offering autonomous capabilities, they

also present various new challenges to systems engineers. Particular challenges originating from each of the three founding domains might affect and amplify each other for SoAS. For example, an unexpected result of an AI/ML algorithm may provide an unexpected behavior for an autonomous system, leading to an undesirable emergent behavior for the entire SoAS.

B. SYSTEM OF SYSTEMS

Integrating multiple systems and forming an SoS provides new capabilities that are not achievable by a single system and are greater than the sum of all capabilities of individual systems [17]. For example, SoSs present interconnected services that lead to improved transportation systems [18], [19], health-care system [20], industrial parks [21], and more effective systems design for various operations including firefighting, marine, and inspection [22], [23], [24], [25], [26].

Maier [27] defined SoS as a collection of individual systems that demonstrates operational and managerial independence, meaning that each system can operate and be managed independently from others [28]. Sage and Cuppan [29] defined SoS as a system that possesses all or majority of the following five characteristics: operational and managerial independence, geographical distribution, emergent behavior, and evolutionary development. Emergent behavior is the result of interactions between systems and is not attributable to any individual system. Desired emergent behaviors are the main reason for constructing SoSs, while undesirable ones need to be tested out in order to prevent adverse consequences [30]. Evolutionary development also means an SoS can be evolved for new purposes by adding/removing/modifying functions/systems.

Furthermore, Boardman and Sauser [17] conducted a literature review and concluded a set of five characteristics to distinguish systems from SoSs: autonomy, belonging, connectivity, diversity, and emergent behaviors. systems practice autonomy as they demonstrate managerial and operational

TABLE 1. SoS Pain Points [31]

<i>SoS Pain Point</i>	<i>Description</i>
<i>Authorities</i>	Multiple system owners need to collaborate and agree on some purposes.
<i>Leadership</i>	More than one organization is involved which leads to a lack of leadership and control.
<i>systems' perspectives</i>	A system that is designed for other purposes needs to be integrated into an SoS to meet a new set of objectives.
<i>Capabilities & Requirements</i>	Meeting the systems' requirements may not address all SoS capability needs.
<i>Autonomy, Interdependencies & Emergence</i>	systems are interdependent within an SoS, meaning that independent changes of one system may lead to unexpected emergent behavior for the SoS.
<i>Testing, validation, and learning</i>	There is not a clear understanding of SoS-level metrics and objectives. SoS evolutionary behavior and systems' interdependencies need to be considered. There might be issues with the funding authority of tests, and tests can be very costly when the SoS is large and diverse.
<i>Principles</i>	The SoS domain is new and more work is needed to identify SoS principles and their applications.

independence, choose to belong to an SoS based on a cost/benefit analysis, and connect to provide increased diversity in SoS capabilities.

Based on a survey's results published by Dahmann [31], seven major pain points, as shown in Table 1, were discovered for implementing systems engineering for SoSs. The pain points include a wide range of different foundational, managerial, and operational challenges a systems engineer may run into throughout the lifecycle of an SoS. This survey also highlights the fact that practicing successful Systems Engineering for SoSs can be very challenging, leading to the development of SoS Engineering to better address SoS pain points [32], [33], [34]. In another study, Honour [35] discusses challenges in verifying and validating SoSs due to evolutionary behavior and changing requirements. All or part of these challenges might also be applicable to SoAS, as an extension of SoS. In addition, they could get worse when impacted by particular issues posed by employing AI-enabled systems.

C. AI/ML

AI is considered to be an enabling factor in engineering autonomy in systems and their operations by replacing and augmenting human operators with computer algorithms [36]. These algorithms enable systems to perceive, learn, think, make decisions, and even act, hence, making them intelligent machines. These AI capabilities assist systems in dealing with uncertain conditions in their operational environment [37]. AI technology also provides the capability of performing tasks

that require a high computational cost in less time than humans, reducing the need for human intervention [38]. Some examples of these tasks include leveraging AI to improve sports technologies such as ball tracking and trajectory [39], industrial predictive maintenance [40], and solar energy forecasting for preventing issues with power grid stability and reliability [41]. Integrating AI-enabled systems into existing operations addresses the needs of today's world such as providing interconnected services like intelligent transportation systems in smart cities [5].

According to Ong and Gupta [42], the three key factors that contributed to AI-enabled systems outperforming humans in various domains in recent years are massive data availability, increased computational capacity, and advancements in AI/ML algorithms. Although AI and ML terms are usually used interchangeably, ML mostly refers to learning methods and can be classified as a subset of AI [36]. Three major types of ML models are supervised, unsupervised, and reinforcement learning [43]. Supervised learning is when the model is trained using labeled data to predict or classify outcomes whereas unsupervised learning primarily focuses on discovering patterns and association rules from unlabeled data. Reinforcement learning is a training method that uses a reward for a desired behavior and punishment for an undesired one [44]. The more accurate these learning methods are, the more effectively the autonomous system can perform a task without the need for human intervention. The focus of this article is ML challenges as it is the predominant form of implementation of AI in autonomous systems, and hence, in SoAS.

ML challenges can be divided into three major categories. The first challenge is concerned with data availability and curation [45]. Systems engineers may have restricted access to data or there may be insufficient data to train an AI algorithm. Moreover, systems engineers may need to preprocess data before using it to train an AI algorithm due to different data types and missing values. This challenge becomes crucial when it impacts the performance of an autonomous system when it uses an AI algorithm that has not been properly trained due to the lack of data availability and curation. The second challenge is that an AI algorithm needs to be resilient against data manipulation or adversarial attacks that lead to misclassification issues [42]. The lack of resilience makes an AI algorithm vulnerable to cyber-attacks and even minor changes in inputs [46]. This can lead to both security breaches and undesirable emergent behaviors for the autonomous system using the algorithm.

Last but not least, some AI algorithms, e.g., Deep Neural Networks, have a black-box nature, making it extremely difficult to explain and interpret input to output mapping [47]. Unknown future data also leads to uncertain outputs from an AI algorithm [43]. It is also difficult to reproduce the obtained results of such an AI algorithm by other algorithms due to a large number of hyperparameters [42]. Unexplainability and irreproducibility of results decrease the trustworthiness in the verification and validation of an AI algorithm, hence, of an

autonomous system [45]. Therefore, every challenge in AI algorithms can lead to serious issues in the SoAS performance that incorporates one or multiple autonomous systems, each having a different AI algorithm.

D. AUTONOMOUS SYSTEMS

An autonomous system can be defined as an engineered system that leverages AI algorithms to perform tasks in its operational environment with less or no human intervention. Researchers realized the necessity to define and quantify system autonomy when AI/ML technologies began to replace human operators in practical systems to carry out tasks. The autonomy levels for unmanned systems framework [15], [16] defines autonomy as “An unmanned system’s own ability of integrated sensing, perceiving, analyzing, communicating, planning, decision-making, and acting/executing, to achieve its goals as assigned.” Adaptation, learning, and failure diagnostics are also other autonomous capabilities mentioned by these two studies [48] and [49]. The advancement of AI/ML in systems has contributed to the increase in their LoA which quantifies the level of autonomous capabilities an unmanned system can perform in a mission without human intervention [15]. Higher LoA means the system provides more autonomous capabilities in its operational environment.

A number of utilization and performance challenges of autonomous systems emerge from the underlying use of AI/ML. These challenges necessitate the advancement of systems engineering approaches when designing AI-enabled systems [45]. The first one is concerned with undesirable emergent behaviors demonstrated through new failure modes and undesirable performances [43]. These behaviors may result from a variety of reasons including limited training data and unknown data that the system will obtain from its future operational environment, resulting in an evolved behavior as the system learns during the operation [50].

The next challenge is concerned with the verification and validation of autonomous systems due to the AI algorithm’s unexplainable and irreproducible results [51]. Trust in a system is achieved by ensuring the system’s performance is validated in the target operational environment. According to the Systems Engineering Handbook [52], validation is the result of comparing the system’s test results against the expected results and deciding on its acceptance based on the degree of compliance. A major concern with autonomous systems is the degree of trust in the system’s future performance [50]. However, the results of an AI algorithm based on unknown data in a future operational environment are not explainable and understandable, leading to a low reliability [43]. Thus, systems engineers are unable to validate the system since they cannot predict the system’s evolving behavior by explaining its AI algorithm’s results.

Last but not least, another crucial challenge is the security breach which is highly related to the challenge of resilience already discussed for AI/ML [1], [42]. Security concerns include the possibility of hacking or other attacks that would

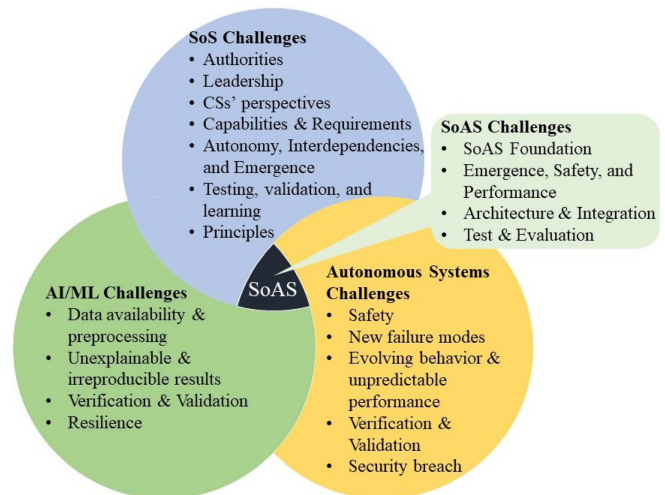


FIGURE 2. SoAS challenges as the intersection of three domains.

allow unauthorized access to sensitive user data [53]. According to a review conducted by Chen et al. [49], major concerns of critics of the use of autonomous vehicles on roads are AI malfunctions, security breaches into users’ privacy, and sharing of tracking data. Thus, the AI/ML challenges have amplified the autonomous systems’ challenges and will, consequently, become crucial for SoAS due to its interconnected network of heterogeneous autonomous systems.

E. SOAS CHALLENGES FOR SYSTEMS ENGINEERS

We are now familiar with the existing particular challenges of each founding domain of SoAS. Fig. 2 demonstrates all the challenges for AI/ML, autonomous systems, and SoS. They may interact with one another and amplify their impact on the SoAS. In this article, we discuss and highlight the interrelationships between these challenges to understand how they result in unique SoAS challenges. We categorize the intersection of challenges originating from the three founding domains, into four overarching SoAS challenges as shown in Fig. 2.

IV. SOAS CHALLENGES

This section describes the identified SoAS challenges as the intersection of particular challenges from the three founding domains. Each following section first describes the SoAS challenge and its importance. Then, the challenge is examined from the perspective of each founding domain to understand how each domain attempts to address it and how it manifests at the SoAS level. The first two challenges are concerned with the conceptual foundations of SoAS, while the other two challenges focus more on the SoAS development.

A. SOAS FOUNDATION

This challenge is concerned with establishing the foundation of SoAS by classifying different SoAS types as well as providing insights into their managerial structures and operational

TABLE 2. Aspects Covered by the Existing Taxonomies in the Founding Domains

Aspects	SoS		AI/ML	Autonomous systems							SoAS
Managerial Autonomy	✓	✓	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Operational Autonomy	N/A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Clear definitions	✓	N/A	N/A	✓	✓	✓	✓	✓	✓	✓	✓
Data exch. (Connectivity)	N/A	✓	✓	✓	✓	N/A	N/A	N/A	N/A	N/A	✓
Multiple systems	✓	✓	✓	✓	✓	Single	Single	Single	Single	Single	✓
Multiple domains	✓	✓	✓	Robotics	Aerospace	Military	Maritime	Automotive	Aviation	✓	✓
References	[27], [54]	[55], [56]	[57]– [65]	[66]	[67]– [73]	[7]	[74]	[69], [75], [76]	[77]	[78]	[6]

behaviors. Each SoAS type can provide a certain level of autonomous capabilities for stakeholders. Similar to SoS, distinguishing and defining different types of SoAS could help with its architecting, integration, and evaluation. Taxonomies can facilitate creating such a classification [79]. A taxonomy provides a hierarchical classification of systems based on a set of concepts. It has been used by many researchers in all three SoAS founding domains. However, none of these taxonomies can address all the SoAS critical aspects that will be discussed in the following. To understand this challenge, we need to first investigate how the existing taxonomies work based on the definition of autonomy in the domains of SoS, AI/ML, and autonomous systems.

The following highlights the semantic ambiguity of the term “autonomy” in the SoAS founding domains. In AI/ML and in autonomous systems, autonomy is mainly focused on operational capabilities previously discussed in Section III, e.g., sensing, analyzing, planning, etc. [15]. In the SoS domain, Darabi et al. [80] and Baldwin and Sauser [81] describe autonomy as the ability of a system to pursue its own purposes. In another study, Delaurentis [56] also defines autonomy as “the degree of control over the entities by an authority or the autonomy granted to the entities.” Thus, autonomy in SoS is concerned with both the managerial and operational aspects. However, they may not be directly applicable to SoAS as they largely ignore the role that LoA plays in SoAS. This is due to the different meanings of operational autonomy for SoS and SoAS. SoS operational autonomy means a system can still operate when detached from the SoS. Whereas operational autonomy for an autonomous system refers to the degree of shared control between its operator and the system based on its AI algorithm. Thus, the SoAS operational autonomy reflects the fact that each system within the SoAS can have its particular LoA, leading to a heterogeneous SoAS with varying LoAs.

The methods for developing taxonomies for SoS, AI/ML, and autonomous systems that have been documented in the literature will be covered in the following. The focus is on highlighting how each founding domain has used a different definition of autonomy to create taxonomies and how they

relate to or impact SoAS. Table 2 summarizes these taxonomies and the different aspects they address.

1) SOS TAXONOMIES

Depending on the managerial independence of systems, Maier [27] defined three SoS types, i.e., directed, collaborative, and virtual. Dahmann and Baldwin [54] also introduced the acknowledged SoS as the fourth type. This taxonomy primarily focuses on the managerial aspect whereas a taxonomy for SoAS needs to also consider the operational perspective of autonomy. Although only four types were discovered, Henshaw et al. [18] believe that it is unlikely that an SoS accurately matches one of the four types. Following this concept, Sauser and Boardman [55] introduced different ranges for each of the five SoS characteristics identified in [17], believing that they are interdependent and different levels of all characteristics link with each other like DNA and create different types of SoS. This taxonomy covers both managerial and operational autonomy.

Similarly, one taxonomy developed by Delaurentis [56] offers a design space that covers the designing of both SoS and SoAS and is based on the degree of connectivity between systems, the degree of systems’ technology, and the degree of managerial and operational autonomy given to systems. Every combination of these degrees results in one type of SoAS. However, they do not clearly distinguish the two types of autonomy in the design space and how varying operational LoAs of autonomous systems can be determined in the design space. Moreover, similar to Sauser and Boardman [55], numerous possible SoAS types can result from their taxonomy, making it very difficult to provide details regarding the characteristics of each SoAS type such as its particular autonomous capabilities.

2) AI/ML TAXONOMIES

The AI domain only focuses on the operational aspect of autonomy and offers four types, i.e., centralized, decentralized, distributed, and hybrid, based on the operational independence of autonomous systems and how they share information. Centralized refers to when the systems transmit

collected observations to a central system, e.g., a ground station, that analyzes and decides on their next move [57], [58], [59], [60], [61]. One significant flaw with this type is that the entire SoAS fails to operate when the central system malfunctions. To address this issue, the decentralized type distributes the workload of the central system among multiple processing units, where each unit controls a small group of systems. In a distributed SoAS, a system is not dependent on a central system and needs to autonomously coordinate with other systems based on the information it receives from them [62], [63]. In a hybrid SoAS, the type is switched between centralized and distributed where high-level decisions are made in a centralized manner while systems are given some degrees of freedom to perform most of the tasks autonomously [64], [65].

This taxonomy has two shortcomings that make it insufficient for understanding SoAS types. First, it does not consider the managerial aspect of autonomy. Second, this taxonomy uses the type of data exchange as the distinguishing metric. In SoAS, a system with a lower LoA might require fewer data from its collaborating systems, whereas the situation might be the opposite for a system with a higher LoA. Each combination of various LoAs results in a certain amount of data exchange. Therefore, each combination can be a particular SoAS type. This leads to a wide range of SoAS types with different characteristics that might fall into the distributed category, however, it is unclear how they differ from one another in this taxonomy. This necessitates that the LoA of systems be taken into account in an SoAS taxonomy in addition to the managerial independence in order for it to accurately describe SoAS types.

3) TAXONOMIES OF AUTONOMOUS SYSTEMS

This domain also focuses only on the operational aspect and offers various taxonomies based on LoA, i.e., the levels of autonomous capabilities. A taxonomy enables a clearer picture of different levels of a system's autonomous capabilities, resulting in a common language for multiple engineering disciplines [70]. According to Gancet et al. [67], a system is more autonomous if it can perform more capabilities independently of human intervention. Comparably, an SoAS can be ranked higher in terms of LoA if its systems are capable of autonomously performing more of the required mission tasks. In order to provide such capabilities, SoAS employs AI cooperation algorithms, making the systems collaborate in performing mission tasks. Table 3 lists some of these capabilities identified in the literature.

As demonstrated in Table 3, the required autonomous capabilities are different depending on the application domains. As a result, there exist various domain-specific taxonomies. There are even different taxonomies within one domain, mostly because of differences in missions and required capabilities. An SoAS may be formed by heterogeneous autonomous systems, each representing a different LoA based on its domain-specific taxonomy. This can lead to problems in cross-domain integration and SoAS testing which typically

TABLE 3. SoAS Capabilities Identified in Various Application Domains

Domain	Mission Setting	Autonomous Capabilities	References
Industry	Loading/Unloading	Task/resource assignment	[82]
	Navigation & Self-localization	Data fusion & map creation	[83], [84]
Military	Search & Attack	Target localization	[58], [59], [85], [86]
		Attacking	[85], [87]
		Formation change	[86]
		Route planning	[88]
		Task/resource assignment	[58], [59], [89]–[93]
Emergency	Disaster Mgt.	Formation change	[94]
	Firefighting	Task/resource assignment	[95]
	Search & Rescue	Target localization	[96], [97]
		Formation change	[98]
		Task/resource assignment	[99]
Surveillance	Targets Search	Target localization	[100]–[106]
		Tracking	[60], [102], [107]
		Data fusion & map creation	[108]–[110]
		Route planning	[100], [111]–[113]
		Task/resource assignment	[57], [114]

do not surface within a specific domain. In addition, Table 2 shows that some of these taxonomies are only applicable to a single system, leading to a limitation in addressing the connectivity aspect. As a result, none of these taxonomies are applicable to address all the required aspects shown in Table 2.

4) SOAS TAXONOMIES

Table 2 summarizes the critical aspects covered in each founding domain when distinguishing different system types. Therefore, except for the “Multiple systems” aspect (which is similar to the “Connectivity” aspect and shows whether the associated taxonomy was defined for a single system or an SoAS), a taxonomy for SoAS must address five major aspects.

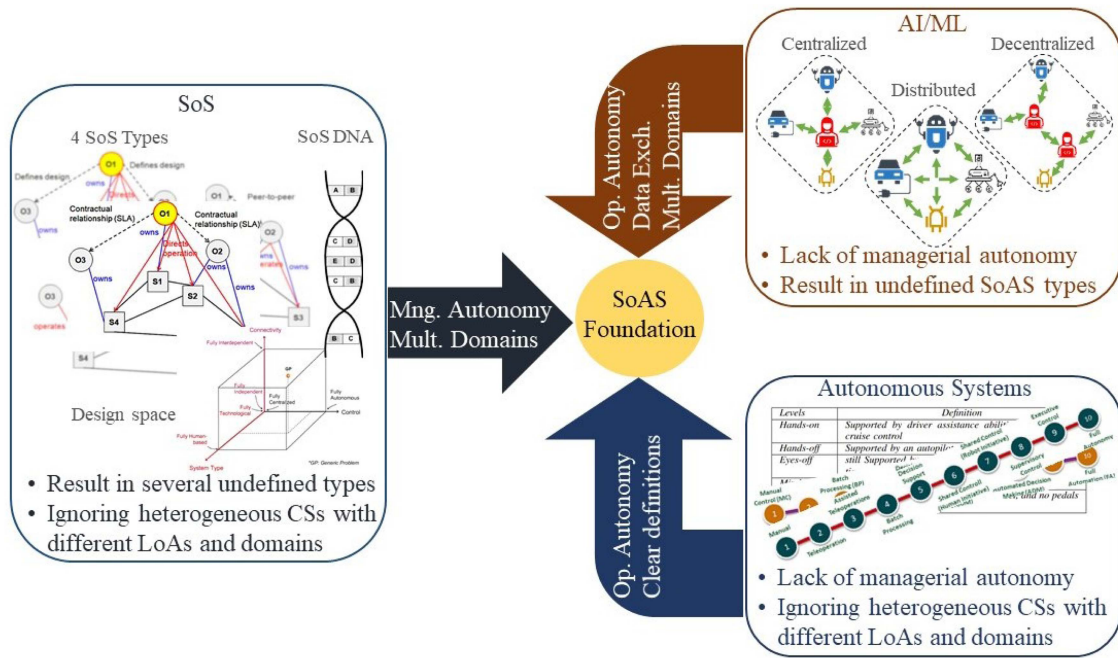


FIGURE 3. Existing taxonomies, their shortcomings, and contributions to enhance the SoAS foundation.

It must consider both the managerial and operational autonomy of systems, clearly distinguish each SoAS type while providing details on its structure and behaviors, must take into account LoA and connectivity between systems, and must be applicable to multiple domains. It can be seen that none of the taxonomies in the founding domains can serve as the one-size-fits-all taxonomy for SoAS.

Among taxonomies proposed for autonomous systems, one taxonomy developed by Endsley and Kaber [78] was found to be applicable to multiple domains due to the use of OODA—Observe, Orient, Decide, and Act [115]—functions. As the number of OODA functions implemented by the system without human intervention increases, the LoA rises. However, this taxonomy was defined for a single system and lacks covering the connectivity and managerial autonomy aspects. Torkjazi and Raz [6] have recently developed a 7-level taxonomy that builds upon OODA as the unifying method for representing SoAS autonomous capabilities in multiple domains. This taxonomy is based on different LoAs of all integrated systems and the amount of information exchanged between them. As Table 2 shows, it addresses all aspects except for the managerial autonomy.

All the aforementioned taxonomies are well-established and have been used in the founding domains to distinguish system types based on a set of characteristics. However, such a taxonomy for SoAS is currently missing within the systems engineering community. Therefore, the remaining challenge is to leverage the existing taxonomies in the founding domains and create a comprehensive SoAS taxonomy addressing all critical aspects listed in Table 2.

5) SUMMARY

Fig. 3 illustrates a summary of the discussion in this section. There is a need for an SoAS taxonomy that:

- 1) provides clear definitions of SoAS types;
- 2) is applicable to multiple domains; and
- 3) distinguishes SoAS types based on managerial autonomy, operational autonomy, and data exchange (i.e., connectivity).

B. EMERGENCE, SAFETY, AND PERFORMANCE

This challenge focuses on undesirable emergent behaviors in SoAS, leading to major concerns such as impacts on safety, operators' performance, and SoAS performance. We will discuss how such emergent behaviors are impacted by complexity factors in each founding domain. This leads to identifying the factors affecting SoAS complexity that lead to SoAS emergence. Then, we will examine how complexity and emergence can positively/negatively impact human interactions with SoAS and its performance.

1) COMPLEXITY SOURCES

a) AI/ML and Autonomous Systems: Complexity in an autonomous system is a combination of the complexity in the AI/ML algorithms and the system itself. Complexity is not solely created within the system, as interactions with people or other systems in the operational environment also result in complexity [116]. A system is complex when it is difficult to completely understand its behavior even by decomposing its structure and studying its parts in isolation [52], [116]. The behavior of an autonomous system is dependent on its

AI/ML algorithm. The uncertainty in its algorithm's output results in unexpected behaviors for the system in its operational environment. These behaviors can be desired as well as undesirable ones. Sections III-C and III-D already discussed the challenges with uncertainty in AI/ML and autonomous systems emergent behaviors.

b) SoS: Complexity in SoS needs to be looked at from both organizational and technical perspectives [31]. The organizational aspect refers to stakeholders, their goals, and leadership. The technical aspect is concerned with independent changes in a system as a result of its autonomy, leading to an increased SoS complexity [31]. Systems interact with one another, hence, independent changes of one system can lead to emergent behaviors for the SoS.

According to Sauser and Boardman [55], the SoS characteristics are associated with different levels and are both independent and interdependent where they influence one another. Darabi et al. [80] formulated the interdependence of autonomy and belonging where the coherence between the purposes of systems and the SoS defines both the levels of belonging (i.e., centralized or decentralized) and autonomy (managerial and operational independence) of each system [55]. As the autonomy of systems increases, the SoS may become more decentralized and need more connectivity between systems to provide information exchange. Diversity in systems also shows the richness of SoS capabilities, and together with connectivity are the determining factors for the level of emergence as a result of interactivity between systems [55].

According to the INCOSE complexity primer [116], "*Complexity results from the diversity, connectivity, interactivity, and adaptivity of a system and its environment.*" Watson et al. [117] also add ten more distinguishing characteristics for a system's complexity such as the system's dynamic states and evolutionary behavior. These factors are the same as the aforementioned SoS characteristics that lead to emergent behaviors. The presence of a higher number of complexity factors in a system or SoS leads to stronger emergence [116], [118].

c) SoAS: In SoAS, the LoA further adds to the dynamics of complexity and corresponding emergent behavior. First, the LoA directly influences the information exchange between systems, making connectivity a function of autonomy in systems. Second, it introduces corresponding complexity from the AI/ML algorithm uncertainty. As a result, LoA is a determining factor for the SoAS complexity. Together, these factors compound with the diversity of systems, where a confounding result will be emergent behaviors that significantly impact human interactions and SoAS operations.

2) IMPACTS OF EMERGENCE

An unanswered question is how emergence affects SoAS and human interactions. For human interactions, the literature led to two important aspects: safety and performance.

a) Safety: Undesirable emergent behaviors at the system level, and consequently, at the SoAS level may lead to adverse outcomes such as life-threatening hazards, hence, need to be detected and eliminated [17]. Henshaw et al. [18] have emphasized the importance of predicting emergent behaviors to reduce the risk of SoS failure in terms of performance and safety.

SoASs are being developed in multiple domains where safety is crucial. Anderson [77] reviewed the challenges of integrating autonomous systems into the National Airspace System. They concluded that safety is the most important challenge due to the operation of autonomous systems in a dynamic air traffic control environment. In their review of applications of SoASs composed of multiple unmanned aerial vehicles for various missions, Skorobogatov et al. [8] listed the safety concern as a disadvantage of these SoASs due to possible collisions between these vehicles.

In the domain of ground transportation, Tien [75] studied the use of Big Data analytics to connect autonomous vehicles and concluded safety as well as data security as two main concerns in such applications. Delaurentis [56] studied the level of shared autonomy between drivers and vehicles in an automotive swarm to ensure safety for the SoS operation. They conclude that the swarm's optimal operating point is when the shared control is distributed as 40% human and 60% vehicle. As another instance, Zhao et al. [119] conducted a safety analysis to analyze the integration of autonomous vehicles into the existing traffic and discovered that the lower reaction time of conventional vehicles is one of the major factors resulting in rear-end collisions.

These are just a few of many studies that highlighted safety concerns in SoASs for daily life applications. The literature is rich in highlighting particular safety considerations in each SoAS founding domain, depending on particular applications such as firefighting, transportation, etc. The current study does not dive into details of safety factors but intends to provide a brief discussion on the importance of safety considerations in SoASs where LoA and uncertainty in AI algorithms can lead to significant undesirable emergent behaviors. Each application domain may require a particular set of safety considerations. However, human-related safety considerations must be the main concern for systems engineers when working on requirements for developing an SoAS [18].

b) Operator's Performance: Although integrating autonomy into SoS operations enhances autonomous capabilities, it may impact human performance. Eight papers in the literature studied the impact of different LoAs on human-machine systems [62], [66], [69], [72], [78], [120], [121], [122]. For instance, Kaber and Endsley [122] conducted an experiment on the performance of 30 human subjects doing a set of cognitive tasks in a computer simulation environment at various LoAs. They concluded that lower LoA resulted in better system performance while hurting the user's performance. Thus, LoA as a complexity factor has significant impacts on both the operator's and the system's performance, consequently influencing the SoAS performance.

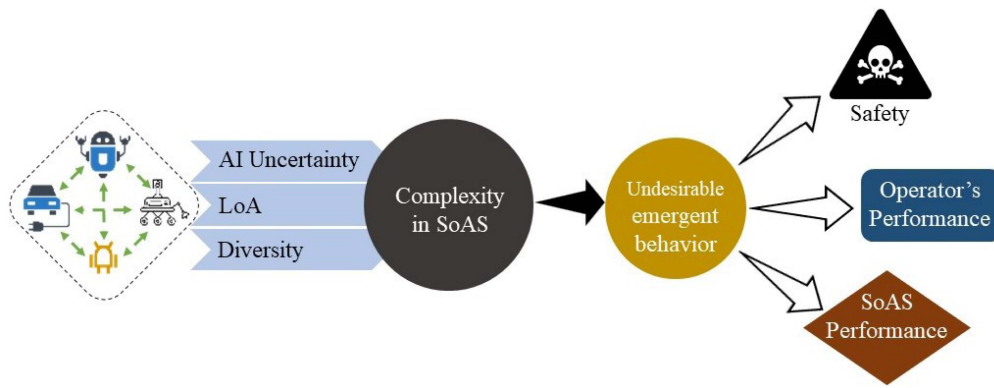


FIGURE 4. Interdependence between SoAS complexity factors, human interaction aspects, and SoAS performance.

c) SoAS Performance: One crucial consideration in SoAS performance is its resilience. Resilience refers to how the SoAS performs within the acceptable level of performance while facing disruptions such as failure in one or more systems [123], [124], [125], [126]. AI/ML uncertainty can lead to unpredicted behaviors at the system level, resulting in disruptions in the SoAS operations. Any disruption can lead to undesirable emergent behaviors for SoAS and reduce its performance. The main goal in resilience studies is identifying a reconfiguration solution without significantly downgrading the system performance [125], [126]. Thus, any increase or decrease in the SoAS complexity factors may also significantly impact its performance.

3) SUMMARY

Fig. 4 summarizes the discussion in this section. Various levels of the three complexity factors can have different impacts on safety and performance. The literature is rich in examining the impact of LoA on safety and performance aspects, whereas the other two complexity factors are ignored. There is a gap in evaluating how all these factors can simultaneously impact safety and performance. Studying this gap will lay the foundation for SoAS test and evaluation efforts. Thus, the main challenge imposed by the three complexity factors is:

- 1) evaluating the individual and interacting impacts of complexity factors on safety, operators' performance, and SoAS performance.

C. ARCHITECTURE AND INTEGRATION

This section highlights the challenges in SoAS architecting and integration. A system's architecture ensures the orchestration of the system's structure and behavior to provide the required capabilities for achieving stakeholders' goals [127]. An architecture establishes the foundation for a system's design and integration. Integration in systems engineering focuses on synthesizing system components to realize the system [128], [129]. In SoS, integration is referred to as the process of synthesizing systems to realize the SoS capabilities [130]. SoS integration is associated with challenges that have their origins in both organizational aspects (e.g., policies

and agreements) and technical aspects (e.g., compatibility and cybersecurity). The same holds for SoAS which is an extension of SoS. However, LoA adds to the SoAS architecting challenges in terms of both aspects. The following examines each of the three founding domains to identify the factors required to be considered for a successful SoAS architecting and integration.

1) ARCHITECTING IN SOS

According to the Wave model proposed by Dahmann et al. [131], two issues make the use of traditional systems engineering quite challenging for SoS development. The first one is technical. When developing an individual system, the systems engineering process focuses on controlling the development environment to optimally meet the system requirements by all components. However, SoS components are the existing systems that may have already been created with a certain number of functions, making it difficult to optimally fulfill the needs of SoS users. The second issue pertains to the SoS organizational aspect which is not required for developing an individual system but must be considered in SoS architecting. For example, lack of control over the development of systems, external changes such as SoS funding, agreements, policies, etc. These issues necessitate the use of SoS engineering instead of systems engineering processes [131]. Therefore, it is essential that systems engineers account for both technical and organizational considerations when developing SoSs. The following introduces some of the proposed SoS architecting approaches and the technical factors to be considered during the SoS integration process.

a) SoS Architecting Approaches: SoS architecting and integration is mostly capability oriented [132]. This concept has resulted in top-down architecting approaches. For example, the Department of Defense developed the Mission Engineering that first analyzes the mission goals and capabilities to be addressed and then develops and evaluates the SoS architecture needed for the mission [133]. As another instance, Giles and Giammarco [134] proposed a mission-based architecting that starts with defining the mission to be accomplished, the different phases of that mission, the operational tasks in each

phase, etc. Franzen et al. [135] also developed a method by using the unified architecture framework (UAF) to break down stakeholders' needs into SoS capabilities and required functions of systems. They, then, imported all this information into an ontology to create the initial SoS design space and decide on the feasible SoS architectures that lead to the desired SoS capabilities in a mission. A limitation of this approach is that capabilities and mission goals only cover the technical aspect while ignoring the organizational aspect of SoS architecting.

To address this need, DeLaurentis and Callaway [136] proposed a lexicon for better understanding an SoS by studying four perspectives, i.e., resources, operations, economics, and policies. Employing this lexicon, Ayyalasomaya-jula et al. [137] formulated an SoS design problem that helps the analyst fully comprehend stakeholders' needs and limitations from different SoS levels and optimize the configuration. Moreover, DeLaurentis [138] leveraged the lexicon and proposes an SoS architecting and analysis approach that has three phases: *Definition* that investigates the status quo of the SoS, *Abstraction* that constructs the main entities of the SoS architecture, and *Implementation* that instantiates the abstracted entities and analyze the SoS by modeling and simulation.

In addition, requirements engineering significantly helps with understanding SoSs. Requirements engineering for a single system simply considers the technical context, while it should also take the entire context—including the social, managerial, political, and organizational dimensions—into account when designing an SoS [139]. To address the need, Keating et al. [139] and Ncube and Lim [140] developed a guideline and discuss research themes for improving the SoS requirements engineering. Such studies give systems engineers insights into approaches for defining different types of requirements that need to be satisfied for SoSs, e.g., requirements for emergent behavior, security, etc.

b) Technical Considerations for SoS Architecture: The technical factors for SoS architecting focus on factors like compatibility, interoperability, data security, etc. One major cause of incompatibility is the systems' failure to communicate and exchange data. This issue originates from the heterogeneity of systems. SoS capabilities can be significantly impacted by architectural constraints enforced by the legacy systems. Legacy systems may impose some architectural constraints that lead to compatibility issues between interfaces, affecting the desired capabilities [141]. This challenge can be addressed by developing a central reference model to ensure semantic understanding by various systems interoperating. For example, Sahin et al. [142] developed an XML-based data communication that is understandable by all systems.

Furthermore, connectivity and data exchange between systems introduce cybersecurity challenges and emphasize the importance of protecting the SoS and systems from unauthorized and illegal access [20], [143]. When developing a system, one important challenge is maintaining the balance between security and operational requirements, meaning that the security requirements should not interrupt the systems' operations [144].

2) ARCHITECTING IN AI/ML AND AUTONOMOUS SYSTEMS

As discussed in Table 2, the AI/ML and autonomous systems literature typically lacks coverage of managerial autonomy. The literature emphasizes the technical factors, such as the algorithm's accuracy and computational efficiency resulting in autonomous capabilities. More autonomous capabilities reflect a more advanced AI algorithm, hence, a higher LoA for the system.

3) ARCHITECTING IN SOAS

SoAS is an extension of SoS, hence, the same organizational and technical factors also matter here. However, these factors are compounded with the LoA originating from AI/ML. Therefore, the impacts of LoA on them must be examined when architecting an SoAS. The following elaborates on these impacts.

a) LoA Impact on Organizational Factors: Organizational factors for SoS include SoS funding, agreements, policies, economics, etc., that may limit certain LoAs of a system in SoAS. The maximum allowed LoA of a system may be restricted by some domain-specific established rules and policies. For example, the military domain restricts the integration of autonomous systems, capable of independent decision-making and action in attack-and-kill missions [7]. Thus, depending on the domain and missions, certain higher LoAs can be restricted while lower LoAs are allowed to be integrated into existing SoSs. This highlights the resulting difference in policies for SoAS architecting due to different LoAs.

b) LoA Impact on Technical Factors: When integrating autonomous systems into an existing SoS, autonomous systems with varying LoAs need to communicate and exchange data with the legacy systems. The major challenge here is to examine interoperability and ensure that systems operate collaboratively. A system with a higher LoA needs to exchange more data with other systems to autonomously decide and act. Thus, depending on the LoA, there might be incompatibility issues in terms of inputs/outputs between the autonomous and legacy systems. LoA also emphasizes cybersecurity considerations. Communications and data exchanges leave the SoAS open to outside threats if cybersecurity concerns are not considered while architecting the SoAS [75]. Improving the interface design is the key to engineering interoperability among systems and between the SoAS and its operational environment, e.g., when integrating autonomous vehicles into smart cities [76]. There are two prominent approaches in the literature, one is based on communication protocols and the other is based on ontology.

Four articles [145], [146], [147], [148] studied possible ways to provide communication and suggested communication protocols to be used among vehicles operating in the same or different environment, such as the air and underwater. For example, Wang et al. [148] reviewed the existing architectures of communication network protocols among unmanned aerial vehicles to help create a reliable

communication network between the vehicles and a ground station. Similarly, Zolich et al. [146] reviewed maritime operations and classified communication protocols required for creating a network of heterogeneous systems to operate in these missions. McGillivray et al. [147] studied the delay/disruption tolerant networking communication protocol to provide reliable communication among vehicles operating in the air, at the surface, and underwater. As another instance, Nagothu et al. [145] proposed protocols for communication between underwater vehicles within themselves and between them and other ones operating in the air or on the ground.

On the contrary, Tolk [149] stated that protocols and data formats refer to the technical challenges of data exchange. However, conceptual challenges such as different data interpretations by different systems are crucial to providing meaningful interoperability. In the SoS Engineering Strategic Research Agenda created by Henshaw et al. [18], they highlighted that many SoS failures are due to issues at the semantic level, meaning that the information exchanged between two systems is interpreted differently by each system. Tolk [149] postulated that integration challenges of autonomous systems originate from the lack of a shared conceptualization between systems. Here, an ontology can create the required shared conceptualization to provide interoperability and common semantics between the systems. For example, an ontology can serve as a reference for simulation interoperability of various systems, each using a different data model [79], [150]. Furthermore, an ontology offers a conceptual alignment that provides a federated representation of the concepts that are common to all systems interacting within the SoAS [151], e.g., the concepts of LoA and autonomous capabilities in different systems.

Both the communication protocol and ontology are complementary approaches and necessary for the successful realization of an SoAS. The communication protocol can be considered as an approach that provides a myopic view of the interface between two systems whereas the ontology provides a holistic view across the SoAS. Standardization bodies are attempting to reconcile the ontology with communication protocols to ensure interoperability within a specific domain. For example, there exist various standardization bodies in different domains (e.g., SAE, European Organization for Civil Aviation Equipment (EUROCAE), etc.), each providing domain-specific standards to ensure interoperability between systems. In the case of an SoAS composed of heterogeneous systems, achieving a shared conceptualization becomes the issue of a consensus and alignment between the existing standards.

4) SUMMARY

The LoA exacerbates both organizational and technical challenges for SoAS architecting. Understanding and addressing these impacts during the SoAS development is crucial to ensure interoperability and compliance with the policies. The existing SoS architecting approaches provide a foundation for

SoAS, however, they need to be reconsidered for the new challenges imposed by the LoA. In summary, the SoAS challenges concluded from this section are as follows:

- 1) understanding how integrating varying LoAs impacts the SoAS organizational and technical factors;
- 2) taking into account the LoA impacts during SoAS development; and
- 3) aligning the existing domain-specific standards to ensure interoperability within the SoAS.

D. TEST AND EVALUATION

This section addresses three main challenges: choosing a comprehensive and efficient testing strategy, selecting the suitable SoAS architecture, and detecting and eliminating undesirable emergent behaviors due to the SoAS complexity factors. The test and evaluation phase during SoAS development assesses the results of addressing the previous three challenges—SoAS foundation, emergent behaviors, and integration.

Dahmann et al. [131] proposed the Wave model for SoS engineering that details the process a systems engineer can use to better address all the key SoS information artifacts. The same applies to SoAS as an extension of SoS. The most important fact about this model is its emphasis on continuous analysis to address the challenges of SoS evolution and systems' incremental development from both technical and organizational aspects. This is the key consideration for SoAS evaluation as the system's LoA may evolve and the effect of this evolution on SoAS can be studied by the Wave model. The organizational aspect addresses the artifacts of risk mitigation, plans, and SoAS-level agreements. The importance of this aspect has already been discussed in the SoAS architecture and integration, Section IV-C1. The technical aspect covers the information artifacts such as technical baselines and performance measures based on different operational contexts [152], [153], [154], [155]. This section reviews the methods, tools, and frameworks proposed in the literature for evaluations in the SoAS founding domains and examines their applicability to address unique challenges for SoAS test and evaluation.

1) TESTING STRATEGY SELECTION

Evaluating an SoAS is much more complicated than just a combination of individual system testings and should include higher level evaluations of interactions among systems and between systems and the operational environment [156]. Due to dynamic interactions among systems, as a result of the AI evolving behavior, SoAS testing needs to be done constantly at both system and SoAS levels, making the testing process complicated and expensive [156], [157]. The crucial challenge here is choosing a testing strategy for SoAS that is both comprehensive and efficient at the same time.

Three studies developed comprehensive testing frameworks for SoS that are also applicable to SoAS. Ferreira et al. [158] reviewed challenges associated with testing autonomous systems, e.g., undesirable emergent behavior and safety, and

proposed a testing framework, i.e., prescriptive and adaptive testing framework (PATFrame). The Department of Defense testing community has used this framework for developing testing strategies and providing a decision support tool. Hess et al. [159] also reviewed real-world testing models used in the Department of Defense operational test beds to identify new testing models and implement them within the PATFrame. Furthermore, Lanus et al. [156] proposed the VTP framework to design a comprehensive test plan that extends testing throughout the system deployment and considers the hierarchy of tests for both system and SoS levels. This framework uses the Vee model and adds the “T” which refers to the testing that happens from the operation stage forward. The “P” also refers to the feedback loop at the end of the VTP framework.

Although these frameworks can significantly assist systems engineers in SoAS evaluations, such comprehensive testing strategies can lead to extra, and sometimes unnecessary, testing costs. The number of systems within the SoAS also adds to the number of tests and required testing resources. There is a need for efficient testing strategies given that testing time and resources might be limited [157]. Efatmaneshnik et al. [157] evaluated various testing architectures and concluded that a modular testing architecture improves the efficiency and effectiveness of system testing. Modular testing manages the combined elements of testing where multiple modules can be tested simultaneously, resulting in less required testing time. However, Efatmaneshnik et al. [157] also stated that modular testing is project-specific, meaning that a testing architecture for one project may not be reusable for another. Thus, the choice of a testing strategy that is comprehensive as well as efficient is still a crucial challenge for SoAS.

2) ARCHITECTURE SELECTION

a) Methods for AI/ML and Individual Autonomous Systems: In the AI/ML domain, test and evaluation are concerned with the choice of the algorithm’s hyperparameters that lead to less uncertainty level in the algorithm’s output. The hyperparameters of an algorithm are trained using a training dataset. Then, the algorithm is evaluated in terms of accuracy, precision, bias, etc. through its implementation on a testing dataset [41], [47]. In order to do this, a dataset is typically split into 70% training and 30% testing sets. There exist various functions and libraries in every programming language that perform simulations to help the analyst choose the combination of hyperparameters that leads to the highest algorithm accuracy, i.e., the algorithm’s optimal operating point.

The uncertainty inherited in AI/ML leads to unexplainable and evolving behavior in an individual autonomous system and may lead to undesirable emergent behaviors at both the system and SoAS levels. Typically, a system is independently trained on its own dataset, while its outputs in SoAS become inputs to other systems. The uncertainty in system outputs may not be included in the training regime of the other systems due to the independent development of systems. Hence,

at the SoAS level, the propagating effect of these uncertainties can compound, leading to emergent behaviors.

b) Methods for SoS: SoS evaluation methods provide an overarching analysis of the SoS complexity, interactions, and interdependence [160]. Fang [161] summarized evaluation methods used for SoS architecture selection such as capability-risk tradeoff analysis, optimization algorithms, agent-based simulation, ML algorithms, and Bayesian Networks. Optimization methods work best when all the needed information, i.e., mission objectives and constraints, is known in advance. Systems engineering researchers employ this approach for identifying the SoS optimal configuration and operating point [4], [162], [163], [164].

Bayesian networks have also been used for architecture evaluation such as evaluating the impact of environmental factors on the performance of a power grid SoS and its systems [165]. In addition, ML algorithms use historical data to assist with predicting SoS performance given the uncertainty in systems [166]. For example, Guariniello et al. [167] employed ML for identifying the relationship between SoS architectural features and SoS-level metrics, leading to generating desired SoS architectures. Last but not least, agent-based simulation has also been used in the literature to find the optimal SoS configuration while taking into account the uncertainty resulting from systems, their interactions, and emergent behaviors [168], [169].

Davendralingam et al. [170] employed a number of aforementioned methods and developed the Analytic Work Bench, which is a suite of evaluation methods and tools for SoS technical analysis. For example, robust portfolio optimization provides a list of SoS architectures that can achieve the desired SoS capabilities [171], systems operational dependence analysis evaluates risk and the SoS resilience based on the systems’ interdependence [172], etc. Some of these methods, such as ML models, Bayesian Networks, and agent-based simulation, are also very beneficial for the SoAS evaluation as they can address uncertainty in SoAS.

c) Methods for SoAS: Similar to the SoS domain, the optimization method has been employed in the literature for offline mission planning, e.g., minimizing mission time and cost, where required information exists [91], [111], [173], [174], [175], [176]. However, dynamic operational environments make this method unable to address the uncertainty challenge in SoAS. To address the need, some researchers developed simulators for both evaluation and training purposes, e.g., automotive industry [177] and military domain [65], [178]. However, creating simulator machines is associated with extra expenses for stakeholders. Alternatively, agent-based simulation provides a cost-effective solution while also dealing with the uncertainty inherited in both individual systems and the SoAS.

While considering the incompatibility issues between legacy and autonomous systems, the architect must also decide on suitable LoAs depending on the needed autonomous capabilities of the SoAS. The choice of LoA impacts the interoperability between autonomous and legacy systems. There

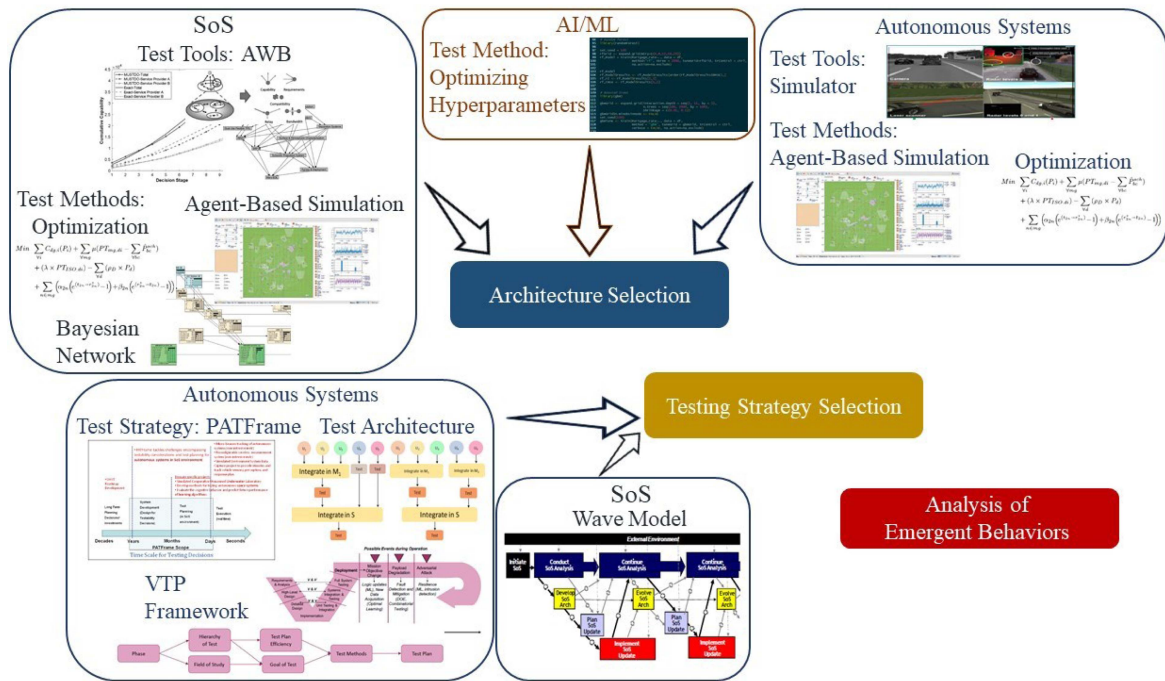


FIGURE 5. Existing test and evaluation methods, tools, and frameworks and their contribution to the SoAS.

exist two approaches that propose various metrics to calculate the required LoA for a single system. The first approach proposed by Huang et al. [15], [16] defined various metrics for mission tasks, the operational environment, and human involvement. Similarly, Parasuraman et al. [179] also defined criteria for measuring the best LoA for a system based on human performance, cost, and reliability. These methods, however, only account for one system at a time, whereas for the SoAS, interactions between multiple systems must also be taken into account.

Simulation-based methods are beneficial in identifying the suitable SoAS architecture in terms of LoA. Three articles study decision-making on system design based on the SoAS performance gap identified in a simulated mission, such as firefighting [180], search and rescue [181], and combat [168], [169]. They make decisions on the autonomous system's design features by mapping system-level metrics to SoAS-level metrics. The idea behind this is that a system with maximum LoA may not even be necessary, as the best system design is the one that delivers the highest SoAS-level metrics [181].

The simulation approach also effectively addresses the uncertainty challenge in SoAS where systems need to constantly exchange information and update their knowledge of the mission [56], [152], [182], [183], [184], [185], [186], [187]. Agent-based simulation has made it simpler and quicker to model collaborating systems and study the SoAS behaviors in various scenarios. Although simulations provide valuable data, the architect must also study possible emergent behaviors, due to uncertainty and evolving behavior of AI/ML algorithms, when deciding on the suitable SoAS architecture.

3) ANALYSIS OF EMERGENT BEHAVIORS

The uncertainty challenge, due to the AI algorithm's accuracy and unexplainable nature, intensifies for the SoAS. The effect of evolution in LoAs of multiple systems compound and can potentially yield undesirable emergent behaviors at the system, and hence, at the SoAS level as a result of their interactions. In addition, the situation might be further exacerbated when also considering the interacting impacts of the other two complexity factors, i.e., LoA and diversity of systems.

Simulations or physical tests provide the data required for studying emergent behaviors. The obtained data needs to be further analyzed by employing other methods, such as statistics and data analysis, to understand emergent behaviors. Identifying the root causes of emergent behaviors can help with predicting and preventing undesirable behaviors in the future. However, the major challenge here is to understand the root causes while accounting for uncertainty in AI algorithms and the individual and interacting impacts of the SoAS complexity factors.

4) SUMMARY

Fig. 5 highlights the key facts on the SoAS test and evaluation discussed in this section. In summary, the main challenges are as follows:

- 1) deciding on a testing strategy that is both efficient and comprehensive;
- 2) selecting the suitable SoAS architecture while accounting for the uncertainty and evolving behaviors due to AI/ML; and
- 3) explaining the root causes of SoAS-level undesirable emergent behaviors.

TABLE 4. Color-Coded Table of the Remaining Research Gaps for SoAS Versus Proposed Research Directions

Challenge	Research Gaps for SoAS	Research Directions
<i>SoAS Foundation</i>	Providing clear definitions of SoAS types	SoAS Foundation Leveraging the existing standards and eliciting experts' experiences in different domains to develop a consensus and create an SoAS ontology that establishes the foundation for SoAS architecting, integration, and evaluation.
	Addressing the multiple domains applicability problem	
	Distinguishing SoAS types based on managerial autonomy, operational autonomy, and data exchange	
<i>Emrg. Sfty., Prf.</i>	Evaluating the individual and interacting impacts of complexity factors on safety, human performance, and SoAS performance	Analysis of LoA impacts on SoAS Updating current systems safety standards to make them applicable to SoAS; Integrating more comprehensive analysis methods with Systems Engineering practices to analyze LoA impacts.
	Aligning the existing domain-specific standards to ensure interoperability within the SoAS	
<i>Architecture & Integration</i>	Understanding how integrating varying LoAs impacts the SoAS organizational and technical factors	Architecture and integration methodology Improving architecting practices by updating existing MBSE methodologies to provide a guide and facilitate the integration of autonomy into existing SoSs.
	Taking into account the LoA impacts during SoAS development	
<i>Test & Evaluation</i>	Deciding on an efficient and comprehensive testing strategy	Test and evaluation methodology Employing new methods, such as ML models, Digital Twins, Bayesian Networks, explainable AI, etc., to develop methods that reduce testing costs, predict emergent behaviors, and identify root causes of such behaviors.
	Selecting the suitable SoAS architecture while accounting for the uncertainty and evolving behaviors due to AI/ML	
	Explaining the root causes of SoAS-level undesirable emergent behaviors	

V. POTENTIAL RESEARCH DIRECTIONS

In this section, we provide potential directions for future SoAS research activities. The topics are classified into four categories, i.e., SoAS foundation, analysis of LoA impacts on SoAS, architecture and integration methodology, and test and evaluation methodology. Table 4 outlines the identified SoAS challenges and their corresponding research directions.

A. SOAS FOUNDATION

As discussed in Section IV-A, there is a need to create a comprehensive SoAS taxonomy that addresses all the five aspects listed in Table 2. Such a taxonomy can improve the SoAS foundation and assist with SoAS architecting, integration, and evaluation. The reason is that it establishes SoAS's various types based on their managerial structures and operational behaviors. This will also help with better decision-making in terms of the suitable testing strategy for an SoAS.

However, creating a practical and pragmatic taxonomy requires a multidisciplinary collaboration across the founding domains of SoAS with subject matter experts from SoS, AI/ML, and autonomous systems. Moreover, SoAS can be composed of heterogeneous systems, each from a different application domain. As discussed in Section IV-C3b, there exist various standardization bodies in different domains (e.g., Society for Automotive Engineers (SAE), European Organization for Civil Aviation Equipment (EUROCAE), etc.), each providing domain-specific standards. Thus, all these knowledge bases must also be used to create a consensus between the existing taxonomies. Such a consensus provides a shared conceptualization and is necessary for representing SoASs

composed of heterogeneous systems from different domains. To create this consensus, *ontologies* may better address this challenge [79].

An ontology offers a shared conceptualization by formulating a conceptual schema within a given domain [150], [188]. Turnitsa and Tolk [151] proposed a method to combine individual ontological representations from separate systems into a federated ontological representation for the SoS within which these systems interact. Similarly, an ontology provides an alignment among the standards from different domains and creates a shared conceptualization for SoAS. An ontology can also identify similarities between different standards, break down their properties, and categorize them with a lower level resolution, resulting in one SoAS ontology.

The resulting SoAS ontology is expected to provide a representation of different SoAS types while employing the existing standards, hence, addressing all the five critical aspects listed in Table 2. This ontology needs to be periodically updated as the field of SoAS is relatively new and requires further development by incorporating experiences gained from real-world projects over time. The developed ontology will serve as a generic ontology from which various domain-specific ontologies can be derived.

B. ANALYSIS OF LOA IMPACTS ON SOAS

As discussed in Sections IV-B, IV-C, and IV-D, LoA can significantly impact the architecting and evaluation of SoAS, and lead to undesirable emergent behaviors at the SoAS level. Failure to understand the LoA impacts may lead to an ineffective integration without achieving the desired capabilities

by the stakeholders. Undesirable emergence due to LoA integration significantly impacts the safety and performance of both the operators and the SoAS. In their study of the SoS Engineering Research Agenda, Henshaw et al. [18] stated that safety research necessitates a multidisciplinary approach in which the human aspects must be incorporated. Systems engineering practices can provide such a multidisciplinary perspective.

Brown et al. [43] emphasized the importance of employing systems engineering practices in designing and developing AI systems to achieve improved safety in their operations. For integrating autonomous systems into the National Airspace System, Anderson [77] concluded that Systems Engineering processes (e.g., requirement and functional analysis, integration, verification, and validation, etc.) are needed to support a balanced approach that provides trust in the ability of the autonomous system to operate safely without endangering or compromising the safety of personnel and properties.

A systems engineering method for developing SoASs ensures taking into account the safety requirements in every phase of the SoAS lifecycle that spans from the SoAS concept to disposal. This helps identify SoAS safety requirements at both organizational and technical levels. The requirements will then be considered for SoAS development and will be used to verify and validate the SoAS. As an example of the organizational aspect, Tien [75] concluded that agencies such as the National Highway Traffic Safety Administration need to publish their first regulations to address safety concerns and crash tests of autonomous vehicles. Such regulations provide the initial requirements needed to be considered during the concept stage of SoAS lifecycle. As a result, the next steps needed for SoAS development will be built upon these requirements to ensure the compliance of the SoAS operations with safety regulations.

Current organizational standards in systems engineering (e.g., DO-178 C for verifying safety-critical software of airborne systems [189]) have established safety levels and how a system must comply with them. However, these standards need to be updated for SoAS to also take into account the new failure modes of AI and autonomous systems discussed in Section III-D. As a result, this could be a potential research direction for various SoAS application domains. As an effort in this field, IEEE has launched the Ethics Certification Program for Autonomous and Intelligent Systems (ECPAIS) to develop certifications for such systems operating in various industries by considering ethical criteria, e.g., accountability, privacy, and algorithmic bias in autonomous and intelligent systems [190]. Their objective is to ensure delivering a more trustworthy system to the users.

Furthermore, current practices for safety analysis, such as failure mode and effects analysis, are more concerned with the technical aspect of safety. Whereas, SoAS is an extension of SoS, necessitating the consideration of both technical and organizational factors such as regulations and policies when analyzing the safety of users, properties in the operational environment, and the SoAS itself. This is true for not only

the safety concerns but all other factors impacted by LoA (e.g., those mentioned for architecting and integration) at both the technical and organizational levels. There exist various comprehensive analysis methods used in other disciplines, such as the SWOT analysis [191] (i.e., *Strength*, *Weakness*, *Opportunity*, and *Threat* in the management studies), that can provide a better analysis of a complex SoAS. As a potential research direction, systems engineers may need to integrate these methods with systems engineering practices to help with understanding the requirements such as safety and security for the SoAS before starting its design, development, and operation. Addressing this need provides the basis for SoAS architecture development and evaluation.

C. ARCHITECTURE AND INTEGRATION METHODOLOGY

As discussed in Section IV-C, LoA significantly impacts the SoAS architecting and integration, adding to the complexity of SoAS architecting and integration. Model-based systems engineering (MBSE) and architecture frameworks provided by the systems engineering domain can handle the added complexity. MBSE aids in providing required information for different groups of stakeholders, modeling and simulation, detecting risks, and supporting the evolutionary development [192]. Architecture frameworks enable an organized MBSE environment for systems engineers.

Architecture frameworks facilitate the application of most of the technical systems engineering processes in the system's life cycle stages of concept, design, and development. Architecture frameworks enable systems engineers to better describe the complexities inherent in the SoAS architecture by creating several views and viewpoints, each of which represents the SoAS to a specific group of stakeholders. The UAF, recently developed by the Object Management Group, provides various viewpoints including the *Security* viewpoint and the *Risk* view that could address the need for including concerns regarding the LoA impacts on SoAS safety and security. Since the introduction of MBSE and the UAF, several researchers proposed approaches to improve their use for architecting SoSs and enterprises, e.g., improving the UAF strategic viewpoint for enterprises [191], incorporating geographical maps into SoS model simulations [193], and integrating data analysis and ML into MBSE [194].

Despite its advantages, the UAF does not provide an architecting methodology to guide systems engineers in modeling the autonomy integration into an existing SoS. Thus, a potential research direction is to empower the UAF by combining it with an established MBSE methodology. An MBSE methodology can provide a step-by-step process to develop different levels of abstraction of the SoAS architecture. Each level of abstraction provides details on both organizational and technical factors for SoAS development. There exist various MBSE methodologies to architect and integrate systems. For instance, INCOSE established the Object-Oriented Systems Engineering Method to guide systems engineers in developing systems [52]. A major contribution to the Systems Engineering community could be updating the current MBSE

methodologies by employing what the UAF provides to account for varying LoAs impacts as well as their modeling in the SoAS architecture. Addressing this need provides the basis for SoAS evaluation which will ensure an effective integration for future operations.

D. TEST AND EVALUATION METHODOLOGY

In the discussion on SoAS test and evaluation, we highlighted three main challenges: testing strategy selection, architecture selection, and analysis of emergent behaviors. In order to decide on a suitable testing strategy, the architect needs information on the SoAS structure and behaviors. As discussed earlier, this information can be provided by establishing an SoAS ontology. Moreover, the literature has already addressed the challenge of architecture selection by employing approaches such as agent-based simulations. Therefore, this section only focuses on the analysis of emergent behaviors and provides research directions that can build upon current practices to address the challenge of understanding the root causes of such behaviors.

Explaining SoAS emergent behaviors necessitates further analysis of the results obtained from real-world operational testing or computer-based simulation. As discussed in Section IV-B3, the literature lacks investigation of the individual and interacting impacts of all three SoAS complexity factors (i.e., LoA, AI uncertainty, diversity of systems) on emergent behaviors. These impacts can be quantified by tracking changes in SoAS-level metrics in response to changes at the system level. Agent-based simulation can help examine changes in the SoAS-level metrics. Furthermore, digital twins can improve the simulation approach by employing real-world data and providing a digital representation of a physical system [194]. However, simultaneously studying multiple factors and testing/simulating every possible SoAS behavior and uncertainty may lead to significant computational cost and time. As a result, field testing and even simulation might not be able to provide a holistic analysis of SoAS emergent behaviors and their root causes. Thus, the main question is how the initial data can be used for analyzing emergent behaviors, aiming to predict and prevent their occurrence.

To address this question in terms of prediction, systems engineers can leverage Digital Twins and pair them with ML models. ML can be used to predict SoS capabilities by understanding decision-making loops at the system level [195]. Such ability can help with predicting emergent behaviors in SoAS-level metrics. Guariniello et al. [69] recommended the use of ML models to identify relationship trends between different LoAs and SoAS-level metrics. Such research studies aid in understanding if integrating higher LoAs causes positive/negative emergent behaviors in SoAS-level metrics. It is worth mentioning that ML models are able to examine complex patterns of relationships within the data while considering multiple factors simultaneously [196]. As a result, one research direction is employing ML models in conjunction with digital twins for understanding not only the impact

of LoA on changes in SoAS-level metrics but also the individual and interacting effects of all three complexity factors on those metrics.

Although ML models are powerful in predicting emergent behaviors, they still may not help identify the behaviors' root causes. This knowledge significantly helps with preventing such behaviors and reducing adverse outputs in future operations of the SoAS. However, the AI unexplainability issue prevents the analyst from understanding how algorithms at the system level translate inputs into outputs at the SoAS level. To address this, there is a need to develop and incorporate explainable AI algorithms. Developing explainable AI has recently been an active research field, particularly after witnessing an increase in using AI-enabled systems that can make decisions that affect humans' lives [197]. However, a larger application across SoAS is missing. Therefore, one research direction is to develop an explainability framework for testing at the SoAS level that expands the explainability of one system and compounds it with the explainability of other systems. The main contribution could be creating such a framework for an SoAS composed of heterogeneous systems and different LoAs.

Bayesian networks can also address the unexplainability problem. Bayesian networks have been used in several studies for causal analysis, probabilistic reasoning, and decision-making under uncertainty [198]. This method has been used to diagnose causes of SoS failures in response to different kinds of disruptions. Bayesian networks can examine various scenarios to study the impact of potential system-level factors and emergent behaviors on SoAS operations. Further research is needed for constructing Bayesian Networks for SoASs while considering several heterogeneous systems, their individual factors, behaviors, and interactions.

VI. CONCLUSION

AI/ML technologies have enabled engineers to integrate autonomy into SoS and build SoAS that provides new autonomous capabilities with limited or no human involvement. SoASs improve effectiveness and efficiency in various military and civilian domains. The SoAS basis lies at the intersection of its founding domains (i.e., SoS, AI/ML, and autonomous systems) where each founding domain has its particular challenges. The primary objective of this article was to study how specific challenges in founding domains can exacerbate their impact and create new challenges for SoAS. This study presented an understanding of these challenges by reviewing 195 published papers in the SoAS founding domains. The reviewed literature consists of theoretical foundations and published results of real-world application case studies. As a result, four unique SoAS challenges were identified that are: 1) SoAS foundation, 2) emergence and its impact on safety and performance, 3) considerations for architecting and integration, and finally, 4) test and evaluation methods and frameworks. The article investigated the methods and tools proposed by researchers and experts in each of the three

founding domains to address related challenges. Then, it highlighted the remaining gaps in the literature and the need for improving existing methods to address the SoAS challenges.

This article also highlighted four potential research directions to assist systems engineers in dealing with SoAS challenges. First is improving the SoAS foundation by providing a consensus on SoAS types that lays the basis for its architecting, integration, and evaluation. The second is concerned with updating current systems safety standards to make them applicable to SoAS and integrating various comprehensive analysis methods with systems engineering practices to analyze LoA impacts on SoAS. The third one focuses on implementing a comprehensive architecting methodology for SoAS while taking into account the LoA impacts on both organizational and technical factors. Finally, further research is needed to improve the SoAS test and evaluation practices by incorporating new methods while also considering undesirable emergent behaviors at the SoAS level.

REFERENCES

- [1] R. Gomes et al., "An interconnected network of UAS as a system-of-systems," in *Proc. IEEE/AIAA 36th Digit. Avionics Syst. Conf.*, 2017, pp. 1–7.
- [2] A. Bürkle, F. Segor, and M. Kollmann, "Towards autonomous micro UAV swarms," *J. Intell. Robot. Syst.*, vol. 61, no. 1, pp. 339–353, 2011.
- [3] A. K. Raz, E. Blasch, R. Cruise, and S. Natarajan, "Enabling autonomy in command and control via game-theoretic models and machine learning with a systems perspective," in *AIAA Scitech 2019 Forum*, Reston, VA, USA: Amer. Inst. Aeronaut. Astronaut., 2019.
- [4] A. K. Marvasti, Y. Fu, S. DorMohammadi, and M. Rais-Rohani, "Optimal operation of active distribution grids: A system of systems framework," *IEEE Trans. Smart Grid*, vol. 5, no. 3, pp. 1228–1237, May 2014.
- [5] INCOSE systems engineering vision 2035, 2021. [Online]. Available: <https://www.incose.org/about-systems-engineering/se-vision-2035>
- [6] M. Torkjazi and A. K. Raz, "A taxonomy for system of autonomous systems," in *Proc. 17th Annu. Syst. Syst. Eng. Conf.*, 2022, pp. 198–203.
- [7] M. T. Ryan and V. Mittal, *Potential for Army Integration of Autonomous Systems by Warfighting Function*. Leavenworth, KS, USA: Army Univ. Press, 2019.
- [8] G. Skorobogatov, C. Barrado, and E. Salami, "Multiple UAV systems: A survey," *Unmanned Syst.*, vol. 08, no. 02, pp. 149–169, 2020.
- [9] E. Honoré-Livemore, A. Dallolio, R. Birkeland, D. D. Langer, C. Haskins, and T. A. Johansen, "MBSE modeling of a SoS with a small satellite and autonomous surface vessels for persistent coastal monitoring," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 156–161.
- [10] J. Zhang, J. Xiong, G. Zhang, F. Gu, and Y. He, "Flooding disaster oriented USV & UAV system development & demonstration," in *Proc. OCEANS 2016 - Shanghai*, 2016, pp. 1–4.
- [11] M. H. Ko, B.-S. Ryuh, K. C. Kim, A. Suprem, and N. P. Mahalik, "Autonomous greenhouse mobile robot driving strategies from system integration perspective: Review and application," *IEEE/ASME Trans. Mechatron.*, vol. 20, no. 4, pp. 1705–1716, Aug. 2015.
- [12] C. Ju and H. Son, "Multiple UAV systems for agricultural applications: Control, implementation, and evaluation," *Electronics*, vol. 7, no. 9, pp. 162–180, 2018.
- [13] M. D. Arango Serna, C. A. Serna Uran, and K. C. Alvarez Uribe, "Collaborative autonomous systems in models of urban logistics," *DYNA*, vol. 79, no. 172, pp. 171–179, 2012.
- [14] A. L. C. Bazzan and F. Klugl, "Agents in traffic and transportation: Exploring autonomy in logistics, management, simulation, and cooperative driving," *Transp. Res. Part C: Emerg. Technol.*, vol. 13, no. 4, pp. 251–254, 2005.
- [15] H.-M. Huang, E. Messina, and J. Albus, "Autonomy levels for unmanned systems (ALFUS) framework, volume II: Framework models version 1.0," NIST Special Publication 1011-II-1.0, 2007.
- [16] H.-M. Huang, J. S. Albus, E. R. Messina, R. C. Wade, and W. English, "Specifying autonomy levels for unmanned systems: Interim report," in *Proc. SPIE Defense Secur. Symp.*, Orlando, FL, USA, 2004. [Online]. Available: <https://www.nist.gov/publications/specifying-autonomy-levels-unmanned-systems-interim-report>
- [17] J. Boardman and B. Sauser, "System of systems—The meaning of of," in *Proc. IEEE/SMC Int. Conf. Syst. Syst. Eng.*, 2006, pp. 386–397.
- [18] M. Henshaw et al., "The systems of systems engineering strategic research agenda," Trans-Atlantic Research and Education Agenda in Systems of Systems (T-AREA-SoS) Project, Aug. 2013.
- [19] S. Henson et al., "Towards a systems of systems engineering EU strategic research agenda," in *Proc. 8th Int. Conf. Syst. Syst. Eng.*, 2013, pp. 99–104.
- [20] N. Wickramasinghe, S. Chalasani, R. V. Boppana, and A. M. Madni, "Healthcare system of systems," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2007, pp. 1–6.
- [21] I. Nikolic and G. P. Dijkema, "Framework for understanding and shaping systems of systems the case of industry and infrastructure development in seaport regions," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2007, pp. 1–6.
- [22] K. Miatliuk, A. Wolniakowski, and P. Kolosowski, "Engineering system of systems conceptual design in theoretical basis of hierarchical systems," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2020, pp. 2850–2856.
- [23] L. Merino, J. R. Martínez-de Dios, and A. Ollero, "Cooperative unmanned aerial systems for fire detection, monitoring, and extinguishing," in *Handbook of Unmanned Aerial Vehicles*, Dordrecht, Netherlands: Springer, 2015, pp. 2693–2722.
- [24] Y. Fan, J. Ma, G. Wang, and T. Li, "Design of a heterogeneous marsupial robotic system composed of an USV and an UAV," in *Proc. 8th Int. Conf. Adv. Comput. Intell.*, 2016, pp. 395–399.
- [25] S. Vemula, S. Marquez, J. D. Avila, T. A. Brouwer, and M. Frye, "A heterogeneous autonomous collaborative system for powerline inspection using human-robotic teaming," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 19–24.
- [26] M. Kahouadji, O. Lakhal, X. Yang, A. Belarouci, and R. Merzouki, "System of robotic systems for crack predictive maintenance," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 197–202.
- [27] M. W. Maier, "Architecting principles for systems-of-systems," *Syst. Eng.*, vol. 1, no. 4, pp. 267–284, 1998.
- [28] M. J. de C. Henshaw, "Systems of systems, cyber-physical systems, the Internet-of-Things...whatever next?," *INSIGHT*, vol. 19, no. 3, pp. 51–54, 2016.
- [29] A. P. Sage and C. D. Cuppan, "On the systems engineering and management of systems of systems and federations of systems," *Inf. Knowl. Syst. Manage.*, vol. 2, no. 4, pp. 325–345, 2001.
- [30] A. Bondavalli, A. Ceccarelli, P. Lollini, L. Montecchi, and M. Mori, "System-of-systems to support mobile safety critical applications: Open challenges and viable solutions," *IEEE Syst. J.*, vol. 12, no. 1, pp. 250–261, Mar. 2018.
- [31] J. Dahmann, "1.4.3 system of systems pain points," in *Proc. INCOSE Int. Symp.*, 2014, pp. 108–121.
- [32] C. Keating et al., "System of systems engineering," *Eng. Manage. J.*, vol. 15, no. 3, pp. 36–45, 2003.
- [33] A. Gorod, A. Fridman, and B. Sauser, "A quantitative approach to analysis of a system of systems operational boundaries," in *Proc. Int. Congr. Ultra Modern Telecommun. Control Syst.*, 2010, pp. 655–661.
- [34] C. Guariniello, A. K. Raz, L. Mockus, and D. DeLaurentis, "Experiences in evolving system-of-systems engineering methodology to address pain points," in *Proc. 14th Annu. Conf. Syst. Syst. Eng.*, 2019, pp. 189–194.
- [35] E. Honour, "Verification and validation issues in systems of systems," *Electron. Proc. Theor. Comput. Sci.*, vol. 133, pp. 2–7, 2013.
- [36] I. H. Sarker, "AI-based modeling: Techniques, applications and research issues towards automation, intelligent and smart systems," *SN Comput. Sci.*, vol. 3, no. 2, 2022, Art. no. 158.
- [37] K. Zhang, Z. Yang, and T. Başar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," in *Handbook of Reinforcement Learning and Control*, Berlin, Germany: Springer, 2021, pp. 321–384.
- [38] B. S. Khan, S. Jangsher, N. Hussain, and M. A. Arafah, "Artificial neural network-based joint mobile relay selection and resource allocation for cooperative communication in heterogeneous network," *IEEE Syst. J.*, vol. 16, no. 4, pp. 5809–5820, Dec. 2022.

- [39] C.-W. Wu et al., "Enhancing fan engagement in a 5G stadium with AI-based technologies and live streaming," *IEEE Syst. J.*, vol. 16, no. 4, pp. 6590–6601, Dec. 2022.
- [40] W. Zhang, D. Yang, and H. Wang, "Data-driven methods for predictive maintenance of industrial equipment: A survey," *IEEE Syst. J.*, vol. 13, no. 3, pp. 2213–2227, Sep. 2019.
- [41] Y. S. Manjili, R. Vega, and M. M. Jamshidi, "Data-analytic-based adaptive solar energy forecasting framework," *IEEE Syst. J.*, vol. 12, no. 1, pp. 285–296, Mar. 2018.
- [42] Y.-S. Ong and A. Gupta, "AIR5: Five pillars of artificial intelligence research," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 3, no. 5, pp. 411–415, Oct. 2019.
- [43] SEBoK, "Artificial intelligence," 2021. [Online]. Available: https://www.sebokwiki.org/w/index.php?title=Artificial_Intelligence&oldid=64868
- [44] Y. Matsuo et al., "Deep learning, reinforcement learning, and world models," *Neural Netw.*, vol. 152, pp. 267–275, 2022.
- [45] A. K. Raz, E. P. Blasch, C. Guariniello, and Z. T. Mian, "An overview of systems engineering challenges for designing AI-enabled aerospace systems," in *AIAA Scitech 2021 Forum*. Reston, VA, USA: American Institute of Aeronautics and Astronautics, 2021.
- [46] D. Hendrycks, N. Carlini, J. Schulman, and J. Steinhardt, "Unsolved problems in ML safety," 2022, *arXiv:2109.13916*. [Online]. Available: <http://arxiv.org/abs/2109.13916>
- [47] A. Parizad and C. Hatziaodoni, "Deep learning algorithms and parallel distributed computing techniques for high-resolution load forecasting applying hyperparameter optimization," *IEEE Syst. J.*, vol. 16, no. 3, pp. 3758–3769, Sep. 2022.
- [48] P. J. Antsaklis and A. Rahnama, "Control and machine intelligence for system autonomy," *J. Intell. Robot. Syst.*, vol. 91, no. 1, pp. 23–34, 2018.
- [49] J. Chen, M. Abbod, and J.-S. Shieh, "Integrations between autonomous systems and modern computing techniques: A mini review," *Sensors*, vol. 19, no. 18, pp. 3897–3914, 2019.
- [50] SEBoK, "Verification and validation of systems in which AI is a key element," 2021. [Online]. Available: https://www.sebokwiki.org/w/index.php?title=Verification_and_Validation_of_Systems_in_Which_AI_is_a_Key_Element&oldid=65244
- [51] A. K. Raz et al., "Explainable AI and robustness based test and evaluation of reinforcement learning," *IEEE Trans. Aerosp. Electron. Syst.*, pp. 1–14, 2024, doi: [10.1109/TAES.2024.3403078](https://doi.org/10.1109/TAES.2024.3403078).
- [52] D. D. Walden et al., Eds. *INCOSE Systems Engineering Handbook*, 5th Ed., 2023.
- [53] J. M. Tien, "Internet of connected ServGoods: Considerations, consequences and concerns," *J. Syst. Sci. Syst. Eng.*, vol. 24, no. 2, pp. 130–167, 2015.
- [54] J. S. Dahmann and K. J. Baldwin, "Understanding the current state of US defense systems of systems and the implications for systems engineering," in *Proc. 2nd Annu. IEEE Syst. Conf.*, 2008, pp. 1–7.
- [55] B. Sausser and J. Boardman, "Taking hold of system of systems management," *Eng. Manage. J.*, vol. 20, no. 4, pp. 3–8, 2008.
- [56] D. A. DeLaurentis, "Appropriate modeling and analysis for systems of systems: Case study synopses using a taxonomy," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2008, pp. 1–6.
- [57] J. Gu, T. Su, Q. Wang, X. Du, and M. Guizani, "Multiple moving targets surveillance based on a cooperative network for multi-UAV," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 82–89, Apr. 2018.
- [58] Y. Jin, A. Minai, and M. Polycarpou, "Cooperative real-time search and task allocation in UAV teams," in *Proc. 42nd IEEE Int. Conf. Decis. Control*, 2003, pp. 7–12.
- [59] Y. Jin, Y. Liao, A. Minai, and M. Polycarpou, "Balancing search and target response in cooperative unmanned aerial vehicle (UAV) teams," *IEEE Trans. Syst., Man, Cybern., Part B (Cybern.)*, vol. 36, no. 3, pp. 571–587, Jun. 2006.
- [60] A. Jaimes, S. Kota, and J. Gomez, "An approach to surveillance an area using swarm of fixed wing and quad-rotor unmanned aerial vehicles UAV(s)," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2008, pp. 1–6.
- [61] P. Angelov, "Collaborative autonomous learning systems," in *Autonomous Learning Systems: From Data Streams to Knowledge in Real-time*, 1st ed. Hoboken, NJ, USA: Wiley, 2013.
- [62] M. L. Cummings, A. Clare, and C. Hart, "The role of human-automation consensus in multiple unmanned vehicle scheduling," *Hum. Factors*, vol. 52, no. 1, pp. 17–27, 2010.
- [63] K. Mori, "Autonomous decentralized systems," in *Wiley Encyclopedia of Computer Science and Engineering*, Hoboken, NJ, USA: Wiley, 2008, pp. 1–6, doi: [10.1002/9780470050118.ecse032](https://doi.org/10.1002/9780470050118.ecse032).
- [64] J. Kvarnström and P. Doherty, "Automated planning for collaborative UAV systems," in *Proc. 11th Int. Conf. Control Autom. Robot. Vis.*, 2010, pp. 1078–1085.
- [65] B. Walter, A. Sannier, D. Reiniers, and J. Oliver, "UAV swarm control: Calculating digital pheromone fields with the GPU," *J. Defense Model. Simul., Appl. Methodol., Technol.*, vol. 3, no. 3, pp. 167–176, 2006.
- [66] M. A. Goodrich, T. W. McLain, J. D. Anderson, J. Sun, and J. W. Crandall, "Managing autonomy in robot teams: Observations from four experiments," in *Proc. 2nd ACM/IEEE Int. Conf. Hum.-Robot Interaction*, 2007, pp. 25–32.
- [67] J. Gancet, G. Hattenberger, R. Alami, and S. Lacroix, "Task planning and control for a multi-UAV system: Architecture and algorithms," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2005, pp. 1017–1022.
- [68] W. Williamson, E. Waltz, J. Mimih, and J. Scrofani, "Autonomy levels for small satellite clusters," in *Proc. IEEE Conf. Cogn. Comput. Aspects Situation Manage.*, 2020, pp. 1–4.
- [69] C. Guariniello, J. Fitch, D. Carpenter, and D. A. DeLaurentis, "Impact of level of autonomy on space systems sizing and system-of-systems metrics," in *Proc. IEEE Aerosp. Conf.*, 2022, pp. 1–10.
- [70] B. Clough, "Metrics, schmetrics! How the heck do you determine a UAV's autonomy anyway," Air Force Res. Lab Wright-Patterson AFB OH, Tech. Rep. ADA515926, 2002, p. 8. [Online]. Available: <https://apps.dtic.mil/sti/citations/ADA515926>
- [71] E. Sholes, "Evolution of a UAV autonomy classification taxonomy," in *Proc. IEEE Aerosp. Conf.*, 2007, pp. 1–16.
- [72] S. Ramchurn, J. Fischer, Y. Ikuno, F. Wu, J. Flann, and A. Waldock, "A study of human-agent collaboration for Multi-UAV task allocation in dynamic environments," in *Proc. Int. Joint Conf. Artif. Intell.*, 2015, pp. 1184–1192.
- [73] M. Woudenbergh, G. Waltensperger, T. Shideler, and J. Franke, "Systems engineering of autonomy: Frameworks for MUM-T architecture," *J. Defense Syst. Inf. Anal. Center*, vol. 7, no. 3, pp. 29–41, 2020.
- [74] T. B. Sheridan and W. L. Verplank, "Human and computer control of undersea teleoperators," Massachusetts Inst. Technol., Cambridge Man-Machine Systems Lab, Tech. Rep. ADA057655, 1978. [Online]. Available: <https://apps.dtic.mil/sti/citations/ADA057655>
- [75] J. M. Tien, "The sputnik of servgoods: Autonomous vehicles," *J. Syst. Sci. Syst. Eng.*, vol. 26, no. 2, pp. 133–162, 2017.
- [76] K. Kuru and W. Khan, "A framework for the synergistic integration of fully autonomous ground vehicles with smart city," *IEEE Access*, vol. 9, pp. 923–948, 2021.
- [77] J. D. Anderson, "Challenges in autonomy: The quest for full autonomy in unmanned flight and the integration of unmanned systems into the NAS," *J. Air Traffic Control*, vol. 53, no. 1, pp. 11–18, 2010.
- [78] M. R. Endsley and D. B. Kaber, "Level of automation effects on performance, situation awareness and workload in a dynamic control task," *Ergonomics*, vol. 42, no. 3, pp. 462–492, 1999.
- [79] A. Tolk, S. Y. Diallo, and C. Turnitsa, "Ontology driven interoperability-M&S applications," *Whitepaper Support I/ITSEC Tutorial*, vol. 2548, pp. 1–14, 2006. [Online]. Available: <http://ontolog.cim3.net/forum/ontolog-forum/2007-01/pdf3oO1b42ltK.pdf>
- [80] H. R. Darabi, M. Mansouri, and A. Gorod, "A mathematical model for formulating interdependence of autonomy and belonging in systems of systems," in *Proc. 7th Int. Conf. Syst. Syst. Eng.*, 2012, pp. 410–415.
- [81] W. C. Baldwin and B. Sausser, "Modeling the characteristics of system of systems," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2009, pp. 1–6.
- [82] L. Zhang, H. Zhong, and S. Y. Nof, "Adaptive fuzzy collaborative task assignment for heterogeneous multirobot systems," *Int. J. Intell. Syst.*, vol. 30, no. 6, pp. 731–762, 2015.
- [83] J. Zhang, R. Liu, K. Yin, Z. Wang, M. Gui, and S. Chen, "Intelligent collaborative localization among air-ground robots for industrial environment perception," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9673–9681, Dec. 2019.
- [84] O. Lakhal, A. Koubeissi, A. Aitouche, C. Sueur, and R. Merzouki, "Autonomous navigation through a system of systems cooperation," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 49–54.
- [85] C. Gao, Z. Zhen, and H. Gong, "A self-organized search and attack algorithm for multiple unmanned aerial vehicles," *Aerosp. Sci. Technol.*, vol. 54, pp. 229–240, 2016.

- [86] J. George, S. P. B., and J. B. Sousa, "Search strategies for multiple UAV search and destroy missions," *J. Intell. Robot. Syst.*, vol. 61, no. 1, pp. 355–367, 2011.
- [87] I. C. Price and G. B. Lamont, "GA directed self-organized search and attack UAV swarms," in *Proc. Winter Simul. Conf.*, 2006, pp. 1307–1315.
- [88] Z. Zhen, P. Zhu, Y. Xue, and Y. Ji, "Distributed intelligent self-organized mission planning of multi-UAV for dynamic targets cooperative search-attack," *Chin. J. Aeronaut.*, vol. 32, no. 12, pp. 2706–2716, 2019.
- [89] H. Wu, H. Li, R. Xiao, and J. Liu, "Modeling and simulation of dynamic ant colony's labor division for task allocation of UAV swarm," *Physica A Statist. Mechan. Appl.*, vol. 491, pp. 127–141, 2018.
- [90] L. Geng, Y. F. Zhang, J. J. Wang, J. Y. H. Fuh, and S. H. Teo, "Cooperative task planning for multiple autonomous UAVs with graph representation and genetic algorithm," in *Proc. 10th IEEE Int. Conf. Control Autom.*, 2013, pp. 394–399.
- [91] W. Xinze, C. Linlin, L. Junshan, and Y. Ning, "Based on PSO algorithm multiple task assignments for cooperating UAVs," in *Proc. Int. Conf. Educ. Inf. Technol.*, 2010, pp. V2-25–V2-28.
- [92] J. Herrmann, "Data-driven metareasoning for collaborative autonomous systems," Digit. Repository Univ. Maryland, Inst. Syst. Res., 2020. [Online]. Available: <https://drum.lib.umd.edu/items/3283bfb-4de6-43d1-8cc0-ea57de1eedfa/full>
- [93] T. Furukawa, F. Bourgault, H. Durrant-Whyte, and G. Dissanayake, "Dynamic allocation and control of coordinated UAVs to engage multiple targets in a time-optimal manner," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2004, pp. 2353–2358.
- [94] E. Budianto et al., "Telecommunication networks coverage area expansion in disaster area using autonomous mobile robots: Hardware and software implementation," in *Proc. Int. Conf. Adv. Comput. Sci. Inf. Syst.*, 2011, pp. 113–118.
- [95] T. Lemaire, R. Alami, and S. Lacroix, "A distributed tasks allocation scheme in multi-UAV context," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2004, pp. 3622–3627.
- [96] S. Waharte, N. Trigoni, and S. Julier, "Coordinated search with a swarm of UAVs," in *Proc. 6th IEEE Annu. Commun. Soc. Conf. Sensor Mesh Ad Hoc Commun. Netw. Workshops*, 2009, pp. 1–3.
- [97] A. Nagaty, C. Thibault, M. Trentini, and H. Li, "Probabilistic cooperative target localization," *IEEE Trans. Autom. Sci. Eng.*, vol. 12, no. 3, pp. 786–794, Jul. 2015.
- [98] J. Scherer et al., "An autonomous Multi-UAV system for search and rescue," in *Proc. 1st Workshop Micro Aerial Veh. Netw. Syst. Appl. Civilian Use*, New York, NY, USA: Association for Computing Machinery, 2015, pp. 33–38.
- [99] E. T. Alotaibi, S. S. Alqefari, and A. Koubaa, "LSAR: Multi-UAV collaboration for search and rescue missions," *IEEE Access*, vol. 7, pp. 55817–55832, 2019.
- [100] A. Sinha, T. Kirubakaran, and Y. Bar-Shalom, "Autonomous search, tracking and classification by multiple cooperative UAVs," in *Signal Processing, Sensor Fusion, and Target Recognition XV*, vol. 6235. Bellingham, WA, USA: SPIE, 2006, pp. 75–86.
- [101] W. Meng et al., "Decentralized control of multi-UAVs for target search, tasking and tracking," *IFAC Proc. Volumes*, vol. 47, no. 3, pp. 10048–10053, 2014.
- [102] M. A. Ma'sum, G. Jati, M. K. Arrofi, A. Wibowo, P. Mursanto, and W. Jatmiko, "Autonomous quadcopter swarm robots for object localization and tracking," in *Proc. MHS*, 2013, pp. 1–6.
- [103] D. Perez, I. Maza, F. Caballero, D. Scarlatti, E. Casado, and A. Ollero, "A ground control station for a multi-UAV surveillance system," *J. Intell. Robot. Syst.*, vol. 69, no. 1, pp. 119–130, 2013.
- [104] J. Chen, W. Zha, Z. Peng, and J. Zhang, "Cooperative area reconnaissance for multi-UAV in dynamic environment," in *Proc. 9th Asian Control Conf.*, 2013, pp. 1–6.
- [105] Y. Zhong, P. Yao, Y. Sun, and J. Yang, "Method of multi-UAVs cooperative search for Markov moving targets," in *Proc. 29th Chin. Control And Decis. Conf.*, 2017, pp. 6783–6789.
- [106] J. J. Acevedo, B. C. Arrue, I. Maza, and A. Ollero, "Distributed cooperation of multiple UAVs for area monitoring missions," in *Motion and Operation Planning of Robotic Systems: Background and Practical Approaches*, Cham, Switzerland: Springer, 2015, pp. 471–494.
- [107] M. Mueller, G. Sharma, N. Smith, and B. Ghanem, "Persistent aerial tracking system for UAVs," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 1562–1569.
- [108] S. Yahyanejad and B. Rinner, "A fast and mobile system for registration of low-altitude visual and thermal aerial images using multiple small-scale UAVs," *ISPRS J. Photogrammetry Remote Sens.*, vol. 104, pp. 189–202, 2015.
- [109] X. Meng, W. Wang, and B. Leong, "SkyStitch: A cooperative multi-UAV-based real-time video surveillance system with stitching," in *Proc. 23rd ACM Int. Conf. Multimedia*, New York, NY, USA: Association for Computing Machinery, 2015, pp. 261–270.
- [110] W. Burgard, M. Moors, D. Fox, R. Simmons, and S. Thrun, "Collaborative multi-robot exploration," in *Proc. ICRA Millennium Conf. IEEE Int. Conf. Robot. Autom. Symposia*, 2000, pp. 476–481.
- [111] G. S. C. Avellar, G. A. S. Pereira, L. C. A. Pimenta, and P. Iscold, "Multi-UAV routing for area coverage and remote sensing with minimum time," *Sensors*, vol. 15, no. 11, pp. 27783–27803, 2015.
- [112] M. Flint, M. Polycarpou, and E. Fernandez-Gaucherand, "Cooperative control for multiple autonomous UAV's searching for targets," in *Proc. 41st IEEE Conf. Decis. Control*, 2002, pp. 2823–2828.
- [113] M. Polycarpou, Y. Yang, and K. Passino, "A cooperative search framework for distributed agents," in *Proc. IEEE Int. Symp. Intell. Control*, 2001, pp. 1–6.
- [114] A. Abel and S. Sukkarieh, "The coordination of multiple autonomous systems using information theoretic political science voting models," in *Proc. IEEE/SMC Int. Conf. Syst. Syst. Eng.*, 2006, pp. 149–154.
- [115] S. E. McIntosh, "The Wingman-philosopher of MiG alley: John Boyd and the OODA loop," *Air Power Hist.*, vol. 58, no. 4, pp. 24–33, 2011.
- [116] INCOSE Complex Systems Working Group, "A Complexity Primer for Systems Engineers Revision 1 2021 White paper," INCOSE-TP-2021-007-01, 2021. [Online]. Available: INCOSE Store
- [117] M. Watson, R. Anway, D. McKinney, L. A. Rosser, and J. MacCarthy, "Appreciative methods applied to the assessment of complex systems," in *Proc. INCOSE Int. Symp.*, 2019, pp. 448–477.
- [118] A. K. Raz, J. Llinas, R. Mittu, and W. Lawless, "Engineering for emergence in information fusion systems: A review of some challenges," in *Proc. 22th Int. Conf. Inf. Fusion*, 2019, pp. 1–8.
- [119] G. Zhao, L. Liu, S. Li, and S. Tighe, "Assessing pavement friction need for safe integration of autonomous vehicles into current road system," *J. Infrastructure Syst.*, vol. 27, no. 2, 2021, Art. no. 04021007.
- [120] A. J. Sengstacken, D. A. DeLaurentis, and M.-R. Akbarzadeh-T, "Fuzzy logic control for shared-autonomy in automotive swarm environment," in *Proc. IEEE Int. Conf. Syst. Man Cybern.*, 2007, pp. 196–201.
- [121] J. Wang, M. Lewis, and P. Scerri, "Cooperating robots for search and rescue," 2006. [Online]. Available: <https://api.semanticscholar.org/CorpusID:10212479>
- [122] D. B. Kaber and M. R. Endsley, "The effects of level of automation and adaptive automation on human performance, situation awareness and workload in a dynamic control task," *Theor. Issues Ergonom. Sci.*, vol. 5, no. 2, pp. 113–153, 2004.
- [123] E. B. Dano, "Resilient system engineering in a multi-UAV system of systems (SoS)," in *Proc. Int. Symp. Syst. Eng.*, 2019, pp. 1–8.
- [124] E. Ordoukhanian and A. M. Madni, "Engineering resilience into multi-UAV systems," *Procedia Comput. Sci.*, vol. 153, pp. 9–16, 2019.
- [125] W. Khalil, R. Merzouki, B. Ould-Bouamama, and H. Haffaf, "Hyper-graph models for system of systems supervision design," *IEEE Trans. Syst., Man, A. Syst. Humans*, vol. 42, no. 4, pp. 1005–1012, Jul. 2012.
- [126] K. Hela, O. Lakhal, B. Conrard, and R. Merzouki, "Formal approach to SoS management design," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 138–143.
- [127] A. K. Raz, C. R. Kenley, and D. A. DeLaurentis, "System architecting and design space characterization," *Syst. Eng.*, vol. 21, no. 3, pp. 227–242, 2018.
- [128] Systems Integration, in *Design and Development of Aircraft Systems*. Hoboken, NJ, USA: Wiley, 2012, pp. 107–137.
- [129] A. Kossiakoff, S. M. Biemer, S. J. Seymour, and D. A. Flanigan, "Systems Integration," in *Systems Engineering Principles and Practice*, 3rd ed. Hoboken, NJ, USA: Wiley, 2020, pp. 485–498.
- [130] T. Mazzuchi and G. Albakri, "7.1.2 system of systems engineering facilitates integration of large scale complex systems," in *Proc. INCOSE Int. Symp.*, 2011, pp. 811–855.
- [131] J. Dahmann, G. Rebovich, J. Lane, R. Lowry, and K. Baldwin, "An implementers' view of systems engineering for systems of systems," in *Proc. IEEE Int. Syst. Conf.*, 2011, pp. 212–217.

- [132] F. Dandashi and M. C. Hause, "UAF for system of systems modeling," in *Proc. 10th Syst. Syst. Eng. Conf.*, 2015, pp. 199–204.
- [133] S. Magnus, "Mission engineering guide," Office of the Under Secretary of Defense for Research and Engineering, Washington, DC, USA, Tech. Rep. 23-S-3518, 2023. [Online]. Available: https://ac.cto.mil/wp-content/uploads/2023/11/MEG_2_Oct2023.pdf
- [134] K. Giles and K. Giammarco, "Mission-based architecture for swarm composability (MASC)," *Procedia Comput. Sci.*, vol. 114, pp. 57–64, 2017.
- [135] L. Knöös Franzén, I. Staack, P. Krus, C. Jouannet, and K. Amadori, "A breakdown of system of systems needs using architecture frameworks, ontologies and description logic reasoning," *Aerospace*, vol. 8, no. 4, pp. 118–144, 2021.
- [136] D. DeLaurentis and R. K. Callaway, "A system-of-systems perspective for public policy decisions," *Rev. Policy Res.*, vol. 21, no. 6, pp. 829–837, 2004.
- [137] S. Ayyalasomayajula, D. Fry, and D. DeLaurentis, "A framework for formulating design problems in a system-of-systems context," in *Proc. 12th AIAA/ISSMO Multidisciplinary Anal. Optim. Conf.*, American Institute of Aeronautics and Astronautics, 2008. [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2008-5818>
- [138] D. A. DeLaurentis, "Understanding transportation as a system-of-systems design problem," in *System of Systems Engineering*, Reston, VA, USA: American Institute of Aeronautics and Astronautics, 2008, pp. 520–541, doi: [10.1002/9780470403501.ch20](https://doi.org/10.1002/9780470403501.ch20).
- [139] C. B. Keating, J. J. Padilla, and K. Adams, "System of systems engineering requirements: Challenges and guidelines," *Eng. Manage. J.*, vol. 20, no. 4, pp. 24–31, 2008.
- [140] C. Ncube and S. L. Lim, "On systems of systems engineering: A requirements engineering perspective and research agenda," in *Proc. IEEE 26th Int. Requirements Eng. Conf.*, 2018, pp. 112–123.
- [141] A. M. Madni and M. Sievers, "System of systems integration: Key considerations and challenges," *Syst. Eng.*, vol. 17, no. 3, pp. 330–347, 2014.
- [142] F. Sahin, M. Jamshidi, and P. Sridhar, "A discrete event XML based simulation framework for system of systems architectures," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2007, pp. 1–7.
- [143] A. Gorod, R. Gove, B. Sauser, and J. Boardman, "System of systems management: A network management approach," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2007, pp. 1–5.
- [144] M. A. Chauhan, M. A. Babar, and S. Grainger, "Designing a security platform for collaborating autonomous systems—An experience report," in *Proc. IEEE 18th Int. Conf. Softw. Archit. Companion*, 2021, pp. 1–7.
- [145] K. Nagothu, T. Shaneyfelt, P. Benavidez, M. A. Jordens, S. Kota, and M. Jamshidi, "Systems of systems communication for heterogeneous independent operable systems," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2009, pp. 1–6.
- [146] A. Zolich et al., "Survey on communication and networks for autonomous marine systems," *J. Intell. Robot. Syst.*, vol. 95, no. 3, pp. 789–813, 2019.
- [147] P. McGillivray, J. Borges de Sousa, R. Martins, K. Rajan, and F. Leroy, "Integrating autonomous underwater vessels, surface vessels and aircraft as persistent surveillance components of ocean observing studies," in *Proc. IEEE/OES Auton. Underwater Veh.*, 2012, pp. 1–5.
- [148] J. Wang, C. Jiang, Z. Han, Y. Ren, R. G. Maunder, and L. Hanzo, "Taking drones to the next level: Cooperative distributed unmanned-aerial-vehicular networks for small and mini drones," *IEEE Veh. Technol. Mag.*, vol. 12, no. 3, pp. 73–82, Sep. 2017.
- [149] A. Tolk, "Conceptual alignment for simulation interoperability: Lessons learned from 30years of interoperability research," *SIMULATION*, vol. 100, pp. 709–726, 2023.
- [150] C. Turnitsa and A. Tolk, "Evaluation of the C2IEDM as an interoperability-enabling ontology," in *Proc. Fall Simul. Interoperability Workshop*, Jun. 2005, pp. 1–13. [Online]. Available: <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=eb6961dbbc4eb27505a54478e8b39e3ced528705>
- [151] C. D. Turnitsa and A. Tolk, "Federated ontologies supporting a merged worldview for distributed systems," in *Proc. AAAI Fall Symp., Regarding Intell. Distrib. Intell. Syst.*, 2007, pp. 116–119.
- [152] P. Xiong, H. Liu, and Y. Tian, "Mission effectiveness evaluation of manned/unmanned aerial team based on OODA and agent-based simulation," in *Proc. 3rd Int. Conf. Artif. Intell. Virtual Reality*, New York, NY, USA: Association for Computing Machinery, 2019, pp. 31–37.
- [153] J. Li, C. Fu, Y. Chen, K. Yang, and X. Zhang, "An operational efficiency evaluation method for weapon system-of-systems combat networks based on operation loop," in *Proc. 9th Int. Conf. Syst. Syst. Eng.*, 2014, pp. 219–223.
- [154] Z. Shu, Q. Jia, X. Li, and W. Wang, "An OODA loop-based function network modeling and simulation evaluation method for combat system-of-systems," in *Theory, Methodology, Tools and Applications for Modeling and Simulation of Complex Systems*, Singapore: Springer, 2016, pp. 393–402.
- [155] J. W. Crandall and M. L. Cummings, "Identifying predictive metrics for supervisory control of multiple robots," *IEEE Trans. Robot.*, vol. 23, no. 5, pp. 942–951, Oct. 2007.
- [156] E. Lanus et al., "Test and evaluation framework for multi-agent systems of autonomous intelligent agents," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 203–209.
- [157] M. Efatmaneshnik, S. Shoval, and K. Joiner, "System test architecture evaluation: A probabilistic modeling approach," *IEEE Syst. J.*, vol. 13, no. 4, pp. 3651–3662, Dec. 2019.
- [158] S. Ferreira et al., "Unmanned and autonomous systems of systems test and evaluation: Challenges and opportunities," Apr. 2010. [Online]. Available: <https://dspace.mit.edu/handle/1721.1/84044?show=full>
- [159] J. Hess et al., "Normative and descriptive models for test & evaluation of unmanned and autonomous systems of systems," in *Proc. 20th Annu. Int. Symp. Int. Council Syst. Eng.*, 2010, pp. 644–654.
- [160] C. Guariniello, A. K. Raz, Z. Fang, and D. DeLaurentis, "System-of-systems tools and techniques for the analysis of cyber-physical systems," *Syst. Eng.*, vol. 23, no. 4, pp. 480–491, 2020.
- [161] Z. Fang, "System-of-systems architecture selection: A survey of issues, methods, and opportunities," *IEEE Syst. J.*, vol. 16, no. 3, pp. 4768–4779, Sep. 2022.
- [162] Y. Li and C. Dagli, "A system of systems approach to optimize a realtime risk situational awareness system," in *Proc. IEEE 15th Int. Conf. Syst. Syst. Eng.*, 2020, pp. 17–22.
- [163] H. Azarnoush, B. Horan, P. Sridhar, A. M. Madni, and M. Jamshidi, "Towards optimization of a real-world robotic-sensor system of systems," in *Proc. World Autom. Congr.*, 2006, pp. 1–8.
- [164] Z. Shu, W. Wang, and R. Wang, "Design of an optimized architecture for manned and unmanned combat system-of-systems: Formulation and coevolutionary optimization," *IEEE Access*, vol. 6, pp. 52725–52740, 2018.
- [165] B. Collins, S. Doskey, and J. Moreland, "Modeling the convergence of collaborative systems of systems: A quantitative case study," *Syst. Eng.*, vol. 20, no. 4, pp. 357–378, 2017.
- [166] A. K. Raz, P. C. Wood, L. Mockus, and D. A. DeLaurentis, "System of systems uncertainty quantification using machine learning techniques with smart grid application," *Syst. Eng.*, vol. 23, no. 6, pp. 770–782, 2020.
- [167] C. Guariniello, L. Mockus, A. K. Raz, and D. A. DeLaurentis, "Towards intelligent architecting of aerospace system-of-systems," in *Proc. IEEE Aerosp. Conf.*, 2019, pp. 1–11.
- [168] P. T. Biltgen and D. N. Mavris, "Capability-based quantitative technology evaluation for systems-of-systems," in *Proc. IEEE Int. Conf. Syst. Syst. Eng.*, 2007, pp. 1–6.
- [169] P. T. Biltgen, "A methodology for capability-based technology evaluation for systems-of-systems," Ph.D. dissertation, Sch. Aerosp. Eng., Georgia Inst. Technol., 2007. [Online]. Available: <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=88f6d6a2719b8922bac36c947f959d7de3902cc7>
- [170] N. Davendralingam et al., "An analytic workbench perspective to evolution of system of systems architectures," *Procedia Comput. Sci.*, vol. 28, pp. 702–710, 2014.
- [171] N. Davendralingam and D. A. DeLaurentis, "A robust portfolio optimization approach to system of system architectures," *Syst. Eng.*, vol. 18, no. 3, pp. 269–283, 2015.
- [172] C. Guariniello and D. DeLaurentis, "Supporting design via the system operational dependency analysis methodology," *Res. Eng. Des.*, vol. 28, no. 1, pp. 53–69, 2017.

- [173] J. Tian, L. Shen, and Y. Zheng, "Genetic algorithm based approach for multi-UAV cooperative reconnaissance mission planning problem," in *Foundations of Intelligent Systems*, Berlin, Germany: Springer, 2006, pp. 101–110.
- [174] C. Ramirez-Atencia, G. Bello-Organ, M. D. R-Moreno, and D. Camacho, "Performance evaluation of multi-UAV cooperative mission planning models," in *Computational Collective Intelligence*, Cham, Switzerland: Springer International Publishing, 2015, pp. 203–212.
- [175] C. Ramirez-Atencia and D. Camacho, "Constrained multi-objective optimization for multi-UAV planning," *J. Ambient Intell. Humanized Comput.*, vol. 10, no. 6, pp. 2467–2484, 2019.
- [176] D. Pascarella, S. Venticinque, and R. Aversa, "Agent-based design for UAV mission planning," in *Proc. 8th Int. Conf. P2P, Parallel Grid Cloud Internet Comput.*, 2013, pp. 76–83.
- [177] F. Bounini, D. Gingras, V. Lapointe, and D. Gruyer, "Poster: Real-time simulator of collaborative autonomous vehicles," in *Proc. Int. Conf. Adv. Comput. Commun. Inform.*, 2014, pp. 1991–1994.
- [178] D. J. Nowak, I. Price, and G. B. Lamont, "SELF organized UAV swarm planning optimization for search and destroy using swarmfire simulation," in *Proc. Winter Simul. Conf.*, 2007, pp. 1315–1323.
- [179] R. Parasuraman, T. Sheridan, and C. Wickens, "A model for types and levels of human interaction with automation," *IEEE Trans. Syst., Man, A, Syst. Humans*, vol. 30, no. 3, pp. 286–297, May 2000.
- [180] M. L. Bociga, "A system-of-systems perspective for simultaneous UAV sizing and allocation using design of experiments and simulation," M.S. thesis, Dept. Sci. Aeronaut. Astronaut., Purdue Univ., 2011. [Online]. Available: <https://docs.lib.purdue.edu/dissertations/AA11501169>
- [181] Y. Tian, H. Liu, and J. Huang, "Aircraft performance analysis in conceptual design phase based on system-of-systems simulations," *MATEC Web Conf.*, vol. 52, 2016, Art. no. 01001.
- [182] Z. Ruiwen, S. Bifeng, P. Yang, and Y. Qijia, "Improved method for subsystems performance trade-off in system-of-systems oriented design of UAV swarms," *J. Syst. Eng. Electron.*, vol. 30, no. 4, pp. 720–737, 2019.
- [183] R. Zhang, B. Song, Y. Pei, W. Tang, and M. Wang, "Agent-based analysis of multi-UAV area monitoring mission effectiveness," in *Proc. AIAA Model. Simul. Technol. Conf.*, 2017, pp. 1–10. [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2017-3151>
- [184] P. Ranque, D. Freeman, K. Kernstine, D. Lim, E. Garcia, and D. Mavris, "Stochastic agent-based analysis of UAV mission effectiveness," in *Proc. 11th AIAA Aviation Technol. Integr. Oper. Conf.*, 2011, pp. 1–15. [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2011-6956>.
- [185] J. Humann and E. Spero, "Modeling and simulation of multi-UAV, multi-operator surveillance systems," in *Proc. Annu. IEEE Int. Syst. Conf.*, 2018, pp. 1–8.
- [186] H. Pfaender, D. DeLaurentis, and D. Mavris, "An object oriented approach for conceptual design exploration of UAV-based system-of-systems," in *Proc. 2nd AIAA "Unmanned Unlimited" Conf. Workshop Exhibit.*, San Diego, CA, USA: American Institute of Aeronautics and Astronautics, 2003, pp. 1–11. [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2003-6521>.
- [187] D. Luch, "Building Multi-UAV simulation methods," in *Proc. AIAA Model. Simul. Technol. Conf. Exhibit.*, American Institute of Aeronautics and Astronautics, 2002. [Online]. Available: <https://arc.aiaa.org/doi/abs/10.2514/6.2002-4977>
- [188] M. Zhang, X. Tang, Q. Bai, and J. Gu, "Expert discovery and knowledge mining in complex multi-agent systems," *J. Syst. Sci. Syst. Eng.*, vol. 16, no. 2, pp. 222–234, 2007.
- [189] Efficient verification through the DO-178C life cycle handbook, 2012. [Online]. Available: https://www.rapitasystems.com/whitepaper/DO-178C-verification?token=IIX957h3KH83tp-VaIl5gM2cCLWhHdHgv6UUVLQ_ea24
- [190] I. S. Association, "The ethics certification program for autonomous and intelligent systems (ECPAIS)," Sep. 2020. [Online]. Available: <https://standards.ieee.org/wp-content/uploads/import/governance/iccom/IC18-004-ECPAIS.pdf>
- [191] M. Torkjazi, A. J. Davila-Andino, A. Alghamdi, and A. K. Zaidi, "UAF strategic planning for enterprises," *IEEE Access*, vol. 10, pp. 123549–123559, 2022.
- [192] D. DeLaurentis, A. Raz, and C. Guariniello, "MBSE for system-of-systems," in *Handbook of Model-Based Systems Engineering*, Cham, Switzerland: Springer, 2020, pp. 1–29.
- [193] Y.-M. Baek, E. Cho, Y.-J. Shin, and D.-H. Bae, "A modeling method for representation of geographical information of a system-of-systems," in *Proc. 16th Int. Conf. Syst. Syst. Eng.*, 2021, pp. 7–12.
- [194] J. Lee, A. Alghamdi, and A. K. Zaidi, "Creating a digital twin of an insider threat detection enterprise using model-based systems engineering," in *Proc. IEEE Int. Syst. Conf.*, 2022, pp. 1–7.
- [195] A. Raz, P. Balasubramani, S. Harrington, C. Guariniello, and D. A. DeLaurentis, "System-of-systems acquisition analytics using machine learning techniques," in *Proc. 17th Annu. Acquisition Res. Symp.*, 2020. [Online]. Available: <https://www.dair.nps.edu/handle/123456789/4235>
- [196] M. A. Farahani, M. R. McCormick, R. Harik, and T. Wuest, "Time-series classification in smart manufacturing systems: An experimental evaluation of state-of-the-art machine learning algorithms," *Robot. Comput.-Integr. Manuf.*, vol. 91, Feb. 2025, Art. no. 102839, doi: [10.1016/j.rcim.2024.102839](https://doi.org/10.1016/j.rcim.2024.102839).
- [197] A. Barredo Arrieta et al., "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, 2020.
- [198] N. U. Ibne Hossain, M. Nagahi, R. Jaradat, C. Shah, R. Buchanan, and M. Hamilton, "Modeling and assessing cyber resilience of smart grid using Bayesian network-based approach: A system of systems problem," *J. Comput. Des. Eng.*, vol. 7, no. 3, pp. 352–366, 2020.



MOHAMMADREZA TORKJAZI (Member, IEEE) received the M.Sc. degree in industrial engineering from West Virginia University, Morgantown, WV, USA, in 2020. He is currently working toward the Ph.D. degree in systems engineering and operations research with George Mason University, Fairfax, VA, USA.

His research interests include developing systems engineering methodologies for integrating autonomous systems, as well as architecting and conducting data-driven evaluations of systems of

systems and enterprises.

Mr. Torkjazi is a Member of the International Council on Systems Engineering and is certified as an Associate Systems Engineering Professional by the organization. He is also certified by the Object Management Group as an OMG-Certified SysML Professional (OCSMP) - SysML Model User.



ALI K. RAZ (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Iowa State University, Ames, IA, USA, and the Ph.D. degree in aeronautics and astronautics from Purdue University, West Lafayette, IN, USA.

He is an Assistant Professor with George Mason University in the Systems Engineering and Operations Research Department and an Associate Director of Intelligent Systems and Integration at the CSI Center. His research and teaching interests are in understanding collaborative autonomy and

developing systems engineering methodologies for integrating autonomous systems. He has worked with Purdue University, Naval Postgraduate School, Johns Hopkins University Applied Physics Laboratory (JHU-APL), the United States Missile Defense Agency, and Honeywell Aerospace. He is a Co-Chair of the International Council on Systems Engineering (INCOSSE) Complex Systems Working Group and a Certified Systems Engineering Professional.

Dr. Raz is also a Senior Member of the American Institute for Aeronautics and Astronautics (AIAA).